

Hidden Exactness in Bounded-Factor Stochastic Models

Variance-Mixture Smiles, Leverage Closure, and Phase-Gram Stationarity

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Bounded stochastic factors are often introduced by transforming an unconstrained driver, but tractability is usually assessed one formula at a time: an exact marginal may survive while first passage, integrated functionals, leverage, or matrix admissibility does not. This paper studies the exactness frontier of the bounded coordinate $\rho_t = X_t/\sqrt{X_t^2 + c^2}$ and of a related Gaussian-angle Gram process. Three results organize the analysis. First, for every centered Gaussian scale mixture with random total standard deviation S , the Bachelier implied standard deviation satisfies $I(0) = E[S]$ and $I''(0) = E[S^{-1}] - E[S]^{-1}$ whenever the inverse moment is finite. For the affine correlation clock $V_T = \beta T + \alpha A_T$, $A_T = \int_0^T \rho_s ds$, this yields an exact at-the-money Jensen wedge: $\rho_{\text{imp}}(0) = E[A_T]/T - \text{Var}(\sqrt{V_T})/(\alpha T)$, and an exact local-curvature sign split between baskets and spreads. Second, Brownian leverage between the factor and asset drivers does not destroy conditional Gaussianity. It produces a path-dependent conditional mean. An explicit Itô gauge converts that mean into one endpoint and one additive functional, while Fourier transformation reduces the terminal characteristic function of a linear asset combination to a one-dimensional complex Feynman–Kac equation; European claims with a valid Fourier representation inherit that closure. Leverage generically creates skewness, while the standard terminal scalar- clock reflection derivation for barriers requires additional structure. Third, independent Brownian triangular angles induce a correlation-matrix process whose unwrapped coordinates diffuse while its matrix marginals converge to a compact stationary pushforward law. The determinant has exact finite-time moments, a stationary product-Beta(1/2, 1/2) law, explicit log-cumulants and a high-dimensional central limit theorem; the survival function of its first singularity time factors into a product of killed-Brownian survival probabilities. Supporting sections correct the scale-identifiability orbit and a nonautonomous Feynman–Kac operator-ordering error, and record exact endpoint-clock and spectrally one-sided Lévy extensions. The paper separates these model-specific results from the classical mixture, evolution-family, hyperspherical, hypoelliptic, and fluctuation-theory ingredients on which they rest.

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1 Introduction

Correlation is simultaneously a bounded state variable, a latent risk factor, and an input to every multi-asset variance calculation. Its stochastic modeling literature contains transform models, Jacobi-type diffusions, Wishart covariance processes, copula dynamics, and local-correlation constructions. These approaches solve different pieces of the same problem: admissibility, mean reversion, finite-time laws, calibration, or derivative pricing (Skintzi and Refenes 2005; Driessen et al. 2009; Fonseca et al. 2007; Ma 2009; Gouriéroux and Sufana 2010; Linders and Schoutens 2014). None of those properties implies the others.

The distinction matters most for path-dependent claims. A bounded transform of a Gaussian driver has an immediate marginal law, but a marginal law does not price an occupation clock, a first-passage claim, or a barrier on a spread. Conversely, a matrix process can remain positive semidefinite by construction while its determinant, boundary behavior, and long-horizon law remain opaque. Exactness is therefore not binary. It is a ledger of which objects survive a modeling operation.

The bounded coordinate studied here is

$$\rho_t = f_c(X_t) = \frac{X_t}{\sqrt{X_t^2 + c^2}}, \quad c > 0. \quad (1)$$

For a Gaussian driver, the map transports transitions and scalar barriers from the real line to $(-1, 1)$. Earlier phases of this research program used that fact to obtain exact correlation transitions, reflection-principle barriers, occupation-clock transforms, and spread/basket prices. The same program introduced a matrix extension: independent Gaussian triangular angles are inserted into a hyperspherical Gram factor, guaranteeing a valid correlation matrix at every time.

The present paper began by auditing where that program claimed exactness was lost. Several boundaries were drawn too broadly. Independence of the factor and asset drivers is not necessary for conditional Gaussianity. Unwrapped Gaussian angles have no invariant probability law on $\mathbb{R}^d$, but the matrix sees only periodic functions of those angles. The standard two-time backward Feynman–Kac theorem does not cease to hold when coefficients depend on calendar time; a time-homogeneous

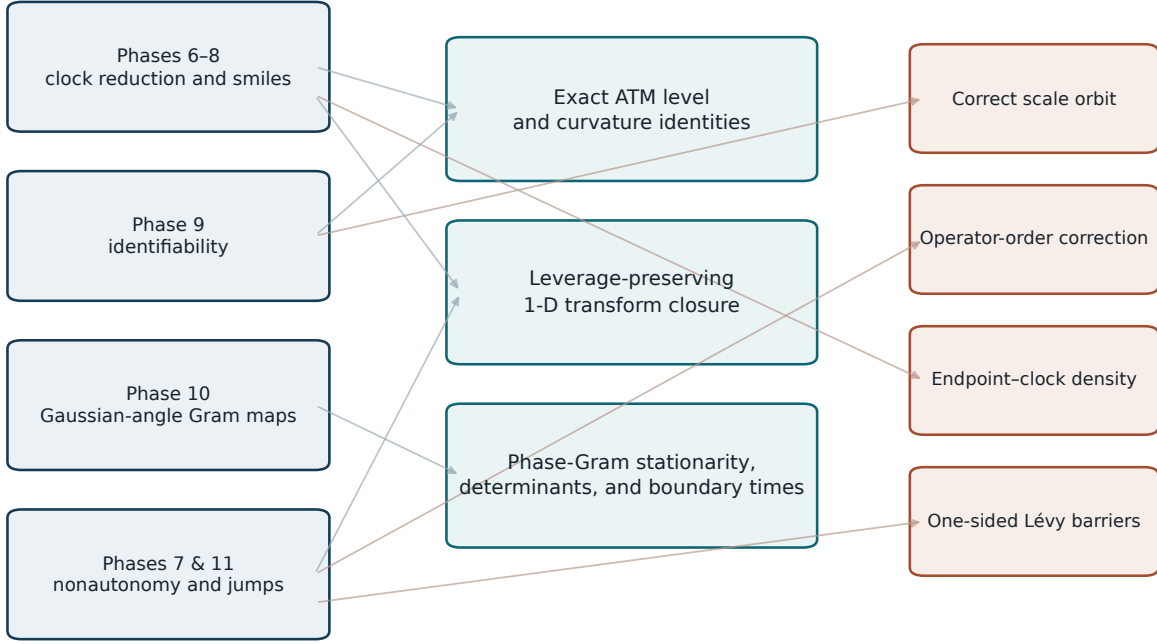


Figure 1: The Phase 14 contribution map. Earlier clock, identifiability, Gram, nonautonomous, and jump results feed three main theorem families and four corrected frontiers.

operator ordering ceases to hold. Reflection fails for generic jump processes, but one-sided Lévy fluctuation identities do not.

The audit also found two concrete errors worth correcting. First, the positive scale orbit of the bounded Gaussian model scales (x_0, c, μ, σ) in the same direction; inverse scaling does not preserve the normalized process. Second, the at-the-money implied correlation of a nondegenerate variance mixture is not the mean correlation clock. The ATM Bachelier price depends on $E[\sqrt{V_T}]$, not $\sqrt{E[V_T]}$.

1.1 Contributions and claim taxonomy

The paper has three principal theorem families and four supporting corrections or exact extensions; Figure 1 displays the dependency map.

1. **Clock-specific smile geometry.** For a centered Gaussian scale mixture, we derive the exact ATM implied-standard-deviation level and curvature. These are compact specializations of established Bachelier stochastic- volatility calculations (Renault and Touzi 1996; Alòs et al. 2024; Alòs and García-Lorite 2025; Alòs and Burés Mogollón 2026). Their use here is the exact correlation-clock corollary: the ATM Jensen wedge and opposite local basket/spread curvature signs.
2. **Bounded-factor leverage closure.** We derive the admissible Brownian leverage geometry, the conditional random-mean Gaussian law, an explicit endpoint-additive Itô gauge, a one-dimensional transform PDE, and a short-time third-moment coefficient. Conditional mixing and Fourier reduction under leverage are established ideas (Willard 1997; Alòs 2006; Heston

- 1993; Das and Langrené 2022); the contribution is their exact specialization and frontier in the bounded-factor system.
3. **Dynamic Phase-Gram analysis.** The hyperspherical map and determinant factorization are classical (Pinheiro and Bates 1996; Rebonato and Jäckel 2000; Rapisarda et al. 2007; Pourahmadi and Wang 2015; Forrester and Zhang 2020). We derive the finite-time determinant moments, compact stationary pushforward, product-Beta equilibrium determinant, and first matrix-singularity law for the specified Brownian-angle dynamics.
 4. **Corrections and exact extensions.** We repair the scale orbit and nonautonomous operator ordering, derive the smooth joint endpoint-clock density through a tilted one-dimensional kernel, and transfer classical one-sided Lévy scale-function barriers through the bounded map.

i Claim categories used throughout

Project-specific result means a theorem or calculation derived for the particular bounded-factor or Brownian-angle model. **Correction** means a previous statement in this research program was mathematically inaccurate or too broad. **Classical specialization** means the theorem follows by applying established machinery to this coordinate. No category by itself is a claim of world priority.

1.2 Related work and the novelty boundary

The normal option formula originates with Bachelier (Bachelier 1900). Modern treatments address stable inversion and the relation between normal and lognormal quotation (Choi et al. 2009, 2022; Jäckel 2017). Arithmetic-Brownian spread pricing and the broader spread-option problem are treated by Poitras (1998) and Carmona and Durrleman (2003).

Normal scale and variance-mean mixtures are classical probability objects (Andrews and Mallows 1974; Barndorff-Nielsen et al. 1982), and stochastic clocks have a long financial history (Clark 1973). In the independent stochastic-volatility setting, the terminal law is an integrated-variance mixture (Hull and White 1987), with symmetric implied-smile consequences (Renault and Touzi 1996). Recent Bachelier work obtains ATM levels, skews, curvatures, and moneyness expansions under broad, sometimes rough, stochastic-volatility assumptions (Alòs et al. 2024; Alòs and García-Lorite 2025; Alòs and Burés Mogollón 2026). Section Section 3 therefore does not claim the general subject of Bachelier curvature. Its distinctive object is the affine correlation clock and the exact conversion of clock dispersion into opposite implied-correlation corrections.

Stochastic correlation is a mature multi-asset modeling problem. Empirical and model-dependent implied-correlation measures are developed by Skintzi and Refenes (2005), Driessen et al. (2009), and Linders and Schoutens (2014). Stochastic-covariance and direct stochastic-correlation models provide analytical or numerical multi-asset pricing (Fonseca et al. 2007; Ma 2009; Escobar et al. 2009, 2016; Gouriéroux and Sufana 2010). The paper does not infer novelty from boundedness alone.

Likewise, leverage need not destroy transform or conditional-mixing methods. Correlated stochastic-volatility models, affine transforms, and random-mean mixing formulas already demonstrate that fact (Romano and Touzi 1997; Willard 1997; Alòs 2006; Heston 1993; Scott 1997; Duffie et al. 2000; Das and Langrené 2022). Barrier reductions may also survive under structures different from the one considered here (Rolloos 2022). Section Section 4 consequently makes a narrower claim: the earlier assertion that leverage destroys conditional Gaussianity in this bounded-factor model is

false, and its correct replacement is an explicit random-mean law and one-dimensional complex Feynman–Kac transform.

The matrix section sits beside a substantial literature on unconstrained covariance parameterizations, random correlation matrices, and determinant-power laws (Joe 2006; Lewandowski et al. 2009; Muirhead 1982; Holmes 1991; Bendel and Mickey 1978; Barnard et al. 2000; Archakov et al. 2024). Dynamic hyperspherical parameters are themselves not new (Bucchieri et al. 2021). The flat-angle invariant law derived below is neither uniform on the ellipsope nor LKJ. That distinction is essential.

1.3 Organization

Section 2 fixes the model and notation. Section 3 develops the exact centered-mixture identities and their correlation-clock consequences. Section 4 treats Brownian leverage. Section 5 derives the matrix laws. Section 6 contains the four supporting corrections and exact extensions. Section 7 assembles the tractability map, limitations, and open problems. Appendices collect calculations that would interrupt the main argument.

2 Bounded-factor framework and conventions

2.1 Probability space and scalar factor

Work on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$ satisfying the usual conditions. Unless another measure is stated, every Brownian motion is standard under $\mathbb{P}$. Fix

$$c > 0, \quad f_c(x) = \frac{x}{\sqrt{x^2 + c^2}}, \quad s_c(x) = \frac{c}{\sqrt{x^2 + c^2}}. \quad (2)$$

Then $f_c^2 + s_c^2 = 1$, f_c is a smooth increasing homeomorphism from $\mathbb{R}$ to $(-1, 1)$, and

$$f_c^{-1}(r) = \frac{cr}{\sqrt{1 - r^2}}, \quad \frac{d}{dr} f_c^{-1}(r) = \frac{c}{(1 - r^2)^{3/2}}. \quad (3)$$

The baseline Gaussian factor is

$$dX_t = \mu dt + \nu dW_t^0, \quad X_0 = x_0, \quad \nu > 0, \quad (4)$$

and the bounded state is $\rho_t = f_c(X_t)$. On every finite interval the state remains strictly inside $(-1, 1)$.

2.2 Two forwards and a linear combination

Let W^1, W^2 be Brownian motions with $d[W^1, W^2]_t = 0$. Under the independence specification used in Section 3, W^0 is independent of (W^1, W^2) . The two Bachelier forward prices satisfy

$$\begin{aligned} dF_{1,t} &= \sigma_1 dW_t^1, \\ dF_{2,t} &= \sigma_2 (\rho_t dW_t^1 + s_c(X_t) dW_t^2). \end{aligned} \quad (5)$$

Thus

$$d[F_1, F_2]_t = \sigma_1 \sigma_2 \rho_t dt.$$

For constants α_1, α_2 , define

$$Z_t = \alpha_1 F_{1,t} + \alpha_2 F_{2,t}, \quad p = \alpha_1 \sigma_1, \quad q = \alpha_2 \sigma_2. \quad (6)$$

The instantaneous variance rate is

$$Q(x) = (p + q f_c(x))^2 + q^2 s_c(x)^2 = p^2 + q^2 + 2pq f_c(x). \quad (7)$$

Write

$$\beta = p^2 + q^2, \quad \alpha = 2pq, \quad A_T = \int_0^T \rho_t dt. \quad (8)$$

Under independence,

$$Z_T \mid \rho_{[0,T]} \sim N(z_0, V_T), \quad V_T = \beta T + \alpha A_T. \quad (9)$$

Since $|A_T| < T$, the total variance lies between $(\beta - |\alpha|)T$ and $(\beta + |\alpha|)T$. We assume it is strictly positive in every result requiring an inverse square-root moment.

2.3 Conventions under leverage

Section 4 permits

$$d[W^0, W^1]_t = \ell_1 dt, \quad d[W^0, W^2]_t = \ell_2 dt, \quad d[W^1, W^2]_t = 0. \quad (10)$$

Whenever a covariance matrix is displayed, the coordinate order is stated. The symbols q and δ are deliberately separated: q is the signed asset loading in Eq. 6, whereas $\delta \geq 0$ is the killing or discount rate in Lévy fluctuation identities.

An equality conditional on X is an equality of regular conditional laws. When a clock J is strictly increasing, the notation $J^{-1}(v) = \inf\{t : J_t \geq v\}$ denotes its time inverse. No inverse-clock representation is imposed when J has flat intervals.

2.4 Phase-Gram construction

For n assets let $d = n(n-1)/2$. For $i > j$, let θ_{ij} be a triangular angle. Define the lower-triangular hyperspherical Gram factor $B(\theta)$ by

$$\begin{aligned} B_{i,k} &= \cos \theta_{ik} \prod_{m < k} \sin \theta_{im}, & k < i, \\ B_{i,i} &= \prod_{m < i} \sin \theta_{im}, & B_{1,1} = 1, \end{aligned} \tag{11}$$

and set

$$R(\theta) = B(\theta)B(\theta)^\top. \tag{12}$$

Every row of B has unit Euclidean norm, so R is symmetric positive semidefinite with unit diagonal. Outside the canonical chart $0 < \theta_{ij} < \pi$, diagonal entries of B can be negative; “Gram factor” is therefore more accurate than “Cholesky factor.”

The dynamic model is

$$d\theta_j(t) = \sigma_j dW_j(t), \quad \sigma_j > 0, \quad j = 1, \dots, d, \tag{13}$$

with independent drivers. Initial angles are deterministic for finite-time product formulas unless stated otherwise. A stationary random initialization is independent of future Brownian increments.

2.5 Notation and hypothesis ledger

Table 1: Standing notation and qualifications.

Symbol or phrase	Meaning and standing condition
f_c, s_c	Bounded coordinate and positive complement in Eq. 2, with $c > 0$.
$S = \sqrt{V}$	Random total standard deviation; $S > 0$ almost surely.
$I(k)$	Bachelier total implied standard deviation at signed moneyness $k = K - z_0$.
$\rho_{\text{imp}}(k)$	Constant correlation matching the mixture call, when the inversion lies in $[-1, 1]$.
$\ell = (\ell_1, \ell_2)$	Primitive Brownian leverage vector; $\ \ell\ ^2 \leq 1$.
Γ	Residual conditional variance rate under leverage.
Π_n	Pushforward of product-uniform angle measure through the Gram map.
$Q_{s,t}$	Two-time tilted evolution family; generator domains share an invariant core when commutators are used.

Symbol or phrase	Meaning and standing condition
K_ξ	Fourier-tilted endpoint kernel, with inversion distributional unless L_ξ^1 decay is proved.
$W^{(\delta)}, Z^{(\delta)}$	Spectrally negative Lévy scale functions under the normalization in Section Section 6.4.

2.6 Reproducible numerical role

All figures and numerical tables are generated by one deterministic script with seed 20260822. The numerical layer checks the exact identities against independent quadrature or simulation. It is not used to infer a theorem. The reference clock parameters are

$$\sigma_1 = 1.5, \quad \sigma_2 = 2.5, \quad x_0 = 0, \quad \mu = 0.3, \quad \nu = 0.8, \quad c = T = 1. \quad (14)$$

3 Exact variance-mixture smile geometry

3.1 Centered Gaussian scale mixtures

Let $S > 0$ be a random variable and suppose that, conditional on S ,

$$Y \mid S \sim N(0, S^2). \quad (15)$$

This is the centered normal scale-mixture structure studied classically in probability (Andrews and Mallows 1974; Barndorff-Nielsen et al. 1982). In the independent-clock model Eq. 9, $Y = Z_T - z_0$ and $S = \sqrt{V_T}$. Conditional normality depends only on the clock; the detailed path producing that clock is irrelevant for a terminal European payoff.

For signed moneyness

$$k = K - z_0,$$

the undiscounted centered Bachelier call is

$$\mathcal{B}(s, k) = s\varphi(k/s) - k\Phi(-k/s), \quad s > 0, \quad (16)$$

where φ and Φ are the standard normal density and distribution function. The mixture call is

$$C(k) = E[\mathcal{B}(S, k)]. \quad (17)$$

The total implied standard deviation $I(k)$ is defined by

$$\mathcal{B}(I(k), k) = C(k). \quad (18)$$

This convention quotes total standard deviation, not annualized normal volatility. The two differ by $\sqrt{T}$.

Lemma 3.1 (Bachelier derivatives and inversion). *For $s > 0$,*

$$\mathcal{B}_s(s, k) = \varphi(k/s) > 0,$$

$$\mathcal{B}_k(s, k) = -\Phi(-k/s), \quad \mathcal{B}_{kk}(s, k) = \frac{\varphi(k/s)}{s}.$$

If $E[S] < \infty$, then for every k the mixture price lies strictly above intrinsic value and Eq. 18 has a unique solution $I(k) > 0$.

Proof. Differentiate Eq. 16 directly, using $\varphi'(x) = -x\varphi(x)$. The terms produced by differentiating $s\varphi(k/s)$ and $-k\Phi(-k/s)$ cancel in the first derivatives. Strict positivity of $\mathcal{B}_s$ gives strict monotonicity in s .

For fixed k , $\mathcal{B}(s, k)$ increases continuously from $(-k)^+$ as $s \downarrow 0$ to infinity as $s \uparrow \infty$. Because $S > 0$ almost surely, $\mathcal{B}(S, k) > (-k)^+$ almost surely, hence $C(k) > (-k)^+$. Integrability follows from the elementary bound $\mathcal{B}(S, k) \leq S/\sqrt{2\pi} + (-k)^+$. The intermediate-value theorem and strict monotonicity complete the proof. $\square$

Normal implied-volatility inversion has its own numerical literature (Choi et al. 2009, 2022; Jäckel 2017). Here inversion is used analytically; no approximation formula is required.

3.2 Exact level and curvature

Theorem 3.1 (Exact centered-mixture ATM identities). *Assume*

$$0 < E[S] < \infty.$$

Then the total implied standard deviation is even and

$$\boxed{I(-k) = I(k), \quad I(0) = E[S].} \tag{19}$$

If additionally

$$E[S^{-1}] < \infty, \tag{20}$$

then I is twice differentiable at zero and

$$\boxed{I''(0) = E[S^{-1}] - \frac{1}{E[S]}.} \tag{21}$$

Consequently,

$$\boxed{(I^2)''(0) = 2(E[S]E[S^{-1}] - 1) \geq 0.} \quad (22)$$

Equality in Eq. 22 holds if and only if S is constant almost surely.

Proof. The Bachelier parity identity in signed-moneyness form is

$$\mathcal{B}(s, -k) = \mathcal{B}(s, k) + k. \quad (23)$$

Taking expectations in Eq. 17,

$$C(-k) = C(k) + k.$$

If $I(k)$ solves Eq. 18, then

$$\mathcal{B}(I(k), -k) = \mathcal{B}(I(k), k) + k = C(-k).$$

Uniqueness from Lemma 3.1 implies $I(-k) = I(k)$.

At zero moneyness,

$$\mathcal{B}(s, 0) = \varphi(0)s.$$

Therefore

$$\varphi(0)I(0) = C(0) = \varphi(0)E[S],$$

which proves the level.

For curvature, Lemma 3.1 gives

$$C'(0) = -\frac{1}{2}.$$

Under Eq. 20, differentiation under the expectation is justified near zero by

$$0 \leq \frac{\varphi(k/S)}{S} \leq \frac{\varphi(0)}{S},$$

so

$$C''(0) = \varphi(0)E[S^{-1}]. \quad (24)$$

Differentiate Eq. 18 once:

$$\mathcal{B}_s(I(k), k)I'(k) + \mathcal{B}_k(I(k), k) = C'(k).$$

At $k = 0$, both $\mathcal{B}_k$ and C' equal $-1/2$, while $\mathcal{B}_s = \varphi(0)$, hence $I'(0) = 0$. A second differentiation gives

$$\mathcal{B}_s I'' + \mathcal{B}_{ss} (I')^2 + 2\mathcal{B}_{sk} I' + \mathcal{B}_{kk} = C''.$$

At zero,

$$\mathcal{B}_{ss} = \mathcal{B}_{sk} = I' = 0, \quad \mathcal{B}_s = \varphi(0), \quad \mathcal{B}_{kk} = \frac{\varphi(0)}{I(0)}.$$

Combine these identities with Eq. 24 and $I(0) = E[S]$ to obtain Eq. 21.

Finally,

$$(I^2)''(0) = 2I(0)I''(0) = 2(E[S]E[S^{-1}] - 1).$$

Cauchy–Schwarz applied to $\sqrt{S}$ and $1/\sqrt{S}$ gives $E[S]E[S^{-1}] \geq 1$. Equality requires $\sqrt{S}$ and $1/\sqrt{S}$ to be linearly dependent almost surely, hence S is constant. $\square$

! Literature boundary

Theorem Theorem 3.1 is an exact finite-horizon specialization of a well-developed Bachelier stochastic-volatility literature. Alòs et al. (2024) treats ATM level, skew, and short-time behavior; Alòs and García-Lorite (2025) gives the exact fixed-maturity Bachelier curvature decomposition; and Alòs and Burés Mogollón (2026) supplies a negative-half-moment moneyness expansion. The compact inverse-moment form is useful here because it exposes the correlation-clock sign algebra immediately; it is not presented as the invention of Bachelier curvature.

The inverse-moment condition is substantive. If S approaches zero too quickly, $C''(0)$ and $I''(0)$ can be infinite. For example, a scale uniformly distributed on $(0, 1)$ has finite mean but infinite $E[S^{-1}]$.

3.3 Correlation inversion and the ATM Jensen wedge

Return to

$$S^2 = V_T = \beta T + \alpha A_T, \quad \alpha \neq 0.$$

The constant-correlation Bachelier comparator has total variance

$$V_T^{\text{const}}(\rho) = \beta T + \alpha T \rho.$$

Define the implied correlation by

$$I(k)^2 = \beta T + \alpha T \rho_{\text{imp}}(k). \tag{25}$$

Because $A_T/T \in (-1, 1)$, the random total standard deviation lies between the two constant-correlation endpoint values. Monotonicity of the Bachelier call in s implies that $I(k)$ lies in the same interval; therefore $\rho_{\text{imp}}(k) \in [-1, 1]$ whenever the endpoint variances are nonnegative. We assume the ATM value is interior when differentiability of the correlation inversion is invoked.

Corollary 3.1 (Exact ATM Jensen wedge and local correlation curvature). *Let $\bar{\rho}_T = E[A_T]/T$. Under the hypotheses of Theorem 3.1,*

$$\rho_{\text{imp}}(0) = \bar{\rho}_T - \frac{\text{Var}(\sqrt{V_T})}{\alpha T}. \quad (26)$$

If $E[V_T^{-1/2}] < \infty$, then

$$\rho''_{\text{imp}}(0) = \frac{2}{\alpha T} \left(E[\sqrt{V_T}] E[V_T^{-1/2}] - 1 \right). \quad (27)$$

For a nonconstant clock, $\rho''_{\text{imp}}(0)$ has the sign of α . Thus a basket ($\alpha > 0$) has positive local correlation curvature and an ATM level below $\bar{\rho}_T$, while a spread ($\alpha < 0$) has negative local correlation curvature and an ATM level above $\bar{\rho}_T$.

Proof. At zero,

$$I(0)^2 = (E[\sqrt{V_T}])^2 = E[V_T] - \text{Var}(\sqrt{V_T}).$$

But

$$E[V_T] = \beta T + \alpha E[A_T].$$

Substitute these identities into Eq. 25 and solve for $\rho_{\text{imp}}(0)$ to obtain Eq. 26.

Since $I'(0) = 0$,

$$\rho''_{\text{imp}}(0) = \frac{(I^2)''(0)}{\alpha T},$$

and Eq. 27 follows from Eq. 22 with $S = \sqrt{V_T}$. Strictness follows from the equality condition in Cauchy–Schwarz. $\square$

The identity repairs a previous mean-clock statement. The error was a Jensen error:

$$E[\sqrt{V_T}] \neq \sqrt{E[V_T]}$$

for a nondegenerate clock. The correction is not an arbitrary approximation; it is exactly $\text{Var}(\sqrt{V_T})$ in total-variance units.

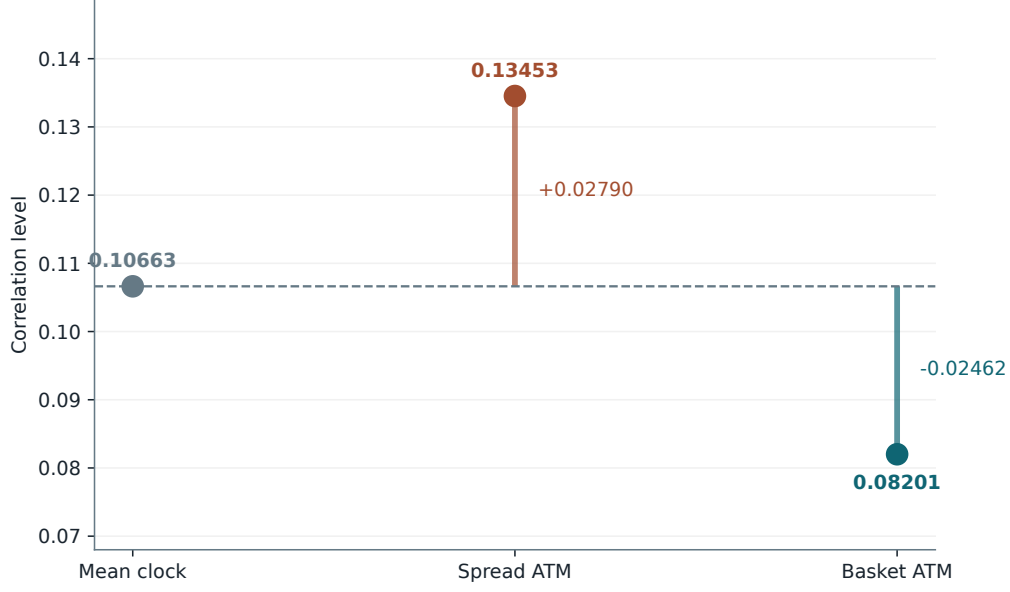


Figure 2: The exact ATM Jensen wedge for the reference clock. The spread lies above the mean clock because its correlation coefficient in total variance is negative; the basket lies below because that coefficient is positive.

3.4 A small-dispersion expansion

The exact wedge is most informative, but a local approximation makes its scale transparent. Write

$$V_T = \bar{v} + \varepsilon, \quad \bar{v} = E[V_T] > 0, \quad E[\varepsilon] = 0.$$

More precisely, consider a family

$$V_T^{(\eta)} = \bar{v} + \eta\xi, \quad E[\xi] = 0, \quad E[|\xi|^5] < \infty,$$

and assume that for all sufficiently small $|\eta|$ there is a fixed $r < 1$ such that $|\eta\xi| \leq r\bar{v}$ almost surely. This uniform separation from zero controls the Taylor remainder. With $\varepsilon = \eta\xi$,

$$\sqrt{\bar{v} + \varepsilon} = \sqrt{\bar{v}} \left[1 + \frac{\varepsilon}{2\bar{v}} - \frac{\varepsilon^2}{8\bar{v}^2} + \frac{\varepsilon^3}{16\bar{v}^3} - \frac{5\varepsilon^4}{128\bar{v}^4} + O(\varepsilon^5) \right].$$

Taking expectations and squaring,

$$(E[\sqrt{V_T}])^2 = \bar{v} - \frac{E[\varepsilon^2]}{4\bar{v}} + \frac{E[\varepsilon^3]}{8\bar{v}^2} + \frac{E[\varepsilon^2]^2 - 5E[\varepsilon^4]}{64\bar{v}^3} + O(\eta^5). \quad (28)$$

For the correlation clock, keep the law of A_T fixed and let $\alpha \rightarrow 0$. Then

$$\bar{v}(\alpha) = \beta T + \alpha E[A_T], \quad \varepsilon = \alpha(A_T - E[A_T]).$$

Because A_T is bounded, the uniform-away-from-zero condition holds for sufficiently small $|\alpha|$. Dividing Eq. 28 by αT gives

$$\rho_{\text{imp}}(0) - \bar{\rho}_T = -\frac{\alpha \text{Var}(A_T)}{4T\bar{v}} + \frac{\alpha^2 E[(A_T - E[A_T])^3]}{8T\bar{v}^2} + O(|\alpha|^3). \quad (29)$$

The leading sign agrees with the exact theorem. The expansion should not be used near a zero-variance boundary; the exact formula remains valid there whenever the relevant moments exist.

3.5 Numerical reconciliation

For Eq. 14, deterministic Gaussian quadrature gives

$$E[A_T] = 0.106628721753, \quad \text{Var}(A_T) = 0.111738386500.$$

The clock-density calculation and direct Bachelier inversion give:

Table 2: Exact reference-clock ATM identities. Values are generated from the bounded clock density and are not fitted.

Case	α	$E[\sqrt{V_T}]$	$\rho_{\text{imp}}(0)$	$\rho_{\text{imp}}(0) - E[A_T]$	$\rho''_{\text{imp}}(0)$
Spread	-7.5	2.73697	0.13453	+0.02790	-0.00799
Basket	+7.5	3.01912	0.08201	-0.02462	+0.00616

The production density uses 160 Fourier modes, 2,001 clock points, and a 1,600-point localized factor grid. Its mean and variance differ from independent Gaussian quadrature by 1.45×10^{-6} and 8.81×10^{-7} , respectively. A separate 192-mode, 2,400-point audit moved the reported ATM correlations by at most 1.2×10^{-6} and the curvatures by less than 2.0×10^{-7} . The table is therefore limited to the digits supported by grid convergence; density normalization alone is not used as a convergence test.

The spread and basket corrections have unequal magnitudes because $\sqrt{\beta T - |\alpha|A_T}$ and $\sqrt{\beta T + |\alpha|A_T}$ have different variances.

Figure 3 illustrates the exact sign mechanism on a transparent two-point family $S \in \{e^{-a}, e^a\}$ with equal probabilities. In that family

$$E[S]E[S^{-1}] - 1 = \sinh^2 a,$$

so the implied-correlation curvatures are $2 \sinh^2 a / (\alpha T)$. The figure is a theorem illustration, not a calibration.

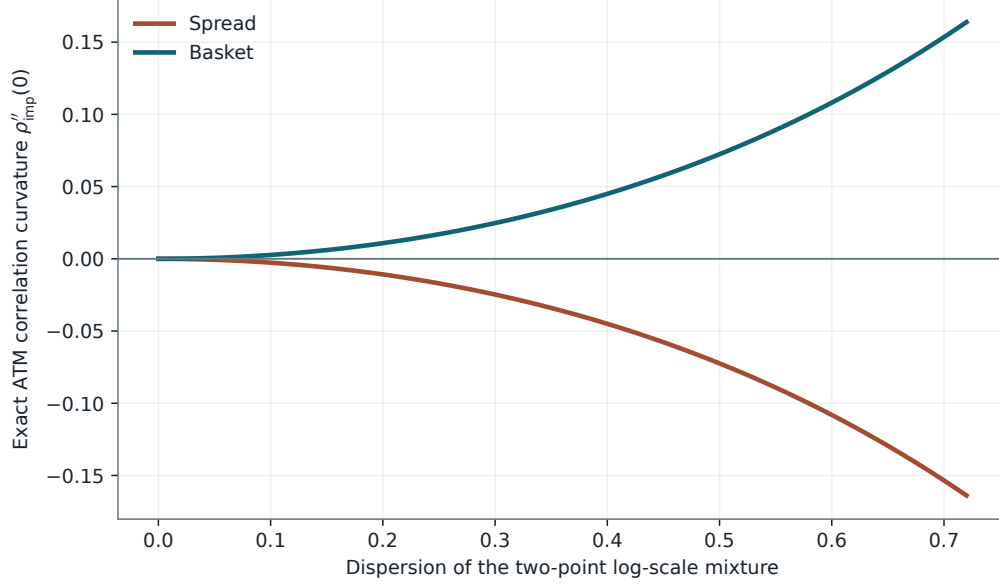


Figure 3: Exact local implied-correlation curvature for an equal-weight two-point log-scale mixture. Dispersion raises the nonnegative curvature numerator; the sign of the correlation curvature is then determined solely by the sign of the variance loading α .

3.6 What the theorem does not imply

The result is local at $k = 0$. It proves:

- exact smile evenness for a centered scale mixture;
- the exact ATM level;
- the existence and sign of finite ATM curvature under an inverse-moment condition; and
- the exact translation into correlation units.

It does **not** prove that curvature retains one sign at every strike, that the wings are monotone, or that every symmetric distribution is a Gaussian scale mixture. Global wing behavior depends on the complete mixing law.

The theorem also fails in its present form when the conditional mean is random. That is precisely what Brownian leverage produces, and it is the subject of Section 4.

4 Leverage and one-dimensional transform closure

4.1 Admissible Brownian geometry

Remove the independence assumption between the factor driver and the primitive asset drivers, retaining

$$d[W^1, W^2]_t = 0, \quad d[W^0, W^1]_t = \ell_1 dt, \quad d[W^0, W^2]_t = \ell_2 dt.$$

In the coordinate order (W^1, W^2, W^0) , the Brownian correlation matrix is

$$\Sigma_W = \begin{pmatrix} 1 & 0 & \ell_1 \\ 0 & 1 & \ell_2 \\ \ell_1 & \ell_2 & 1 \end{pmatrix}. \quad (30)$$

Lemma 4.1 (Admissible primitive leverage region). *The Brownian system Eq. 30 exists if and only if*

$$\boxed{\ell_1^2 + \ell_2^2 \leq 1}. \quad (31)$$

Proof. The one-dimensional principal minors equal one. The nontrivial two-dimensional principal minors are

$$1, \quad 1 - \ell_1^2, \quad 1 - \ell_2^2,$$

and

$$\det \Sigma_W = 1 - \ell_1^2 - \ell_2^2.$$

Nonnegativity of the determinant implies nonnegativity of the two leverage minors. Conversely, nonnegativity of all principal minors is sufficient for a symmetric matrix to be positive semidefinite. $\square$

Pairwise choices inside $[-1, 1]$ are not enough. For example, $(\ell_1, \ell_2) = (0.8, 0.8)$ violates Eq. 31.

Let $\ell = (\ell_1, \ell_2)^\top$. Choose any matrix C such that

$$CC^\top = I_2 - \ell\ell^\top, \quad (32)$$

and let $U = (U^1, U^2)$ be a two-dimensional Brownian motion independent of W^0 . Then

$$\begin{pmatrix} dW_t^1 \\ dW_t^2 \end{pmatrix} = \ell dW_t^0 + C dU_t. \quad (33)$$

When $|\ell_1| < 1$, one explicit choice is

$$\begin{aligned} dW_t^1 &= \ell_1 dW_t^0 + \sqrt{1 - \ell_1^2} dU_t^1, \\ dW_t^2 &= \ell_2 dW_t^0 - \frac{\ell_1 \ell_2}{\sqrt{1 - \ell_1^2}} dU_t^1 + \sqrt{\frac{1 - \ell_1^2 - \ell_2^2}{1 - \ell_1^2}} dU_t^2. \end{aligned} \quad (34)$$

The construction preserves $d[W^1, W^2]_t = 0$. In particular, it preserves the intended instantaneous asset correlation:

$$\text{Corr}(dF_{1,t}, dF_{2,t} \mid \mathcal{F}_t) = f_c(X_t).$$

The correlation between the factor and the second asset return is

$$\text{Corr} \left(dW_t^0, \frac{dF_{2,t}}{\sigma_2} \middle| \mathcal{F}_t \right) = \ell_1 f_c(X_t) + \ell_2 s_c(X_t), \quad (35)$$

so constant leverage against the primitive orthogonal drivers becomes state-dependent leverage against the rotated second asset.

The instantaneous covariance-rate matrix of (F_1, F_2, X) has determinant

$$\sigma_1^2 \sigma_2^2 \nu^2 s_c(X_t)^2 (1 - \ell_1^2 - \ell_2^2) \geq 0. \quad (36)$$

This determinant follows either by direct calculation or by taking the squared determinant of the loading matrix times $\det \Sigma_W$.

4.2 Conditional Gaussianity with a random mean

For the linear combination Eq. 6 define

$$g_1(x) = p + q f_c(x), \quad g_2(x) = q s_c(x), \quad g(x) = \begin{pmatrix} g_1(x) \\ g_2(x) \end{pmatrix}. \quad (37)$$

The component exposed to the factor Brownian motion is

$$h(x) = \ell^\top g(x) = \ell_1(p + q f_c(x)) + \ell_2 q s_c(x). \quad (38)$$

The residual loading and residual variance rate are

$$r(x) = C^\top g(x),$$

$$\Gamma(x) = \|r(x)\|^2 = g(x)^\top (I_2 - \ell \ell^\top) g(x) = Q(x) - h(x)^2 \geq 0. \quad (39)$$

Consequently,

$$dZ_t = h(X_t) dW_t^0 + r(X_t)^\top dU_t. \quad (40)$$

Theorem 4.1 (Leveraged conditional Gaussian law). *Conditional on the entire factor path,*

$$\boxed{Z_T \mid X_{[0,T]} \sim N \left(z_0 + \int_0^T h(X_t) dW_t^0, \int_0^T \Gamma(X_t) dt \right)}. \quad (41)$$

More generally, the conditional process is Gaussian with mean curve

$$m_t(X) = z_0 + \int_0^t h(X_s) dW_s^0$$

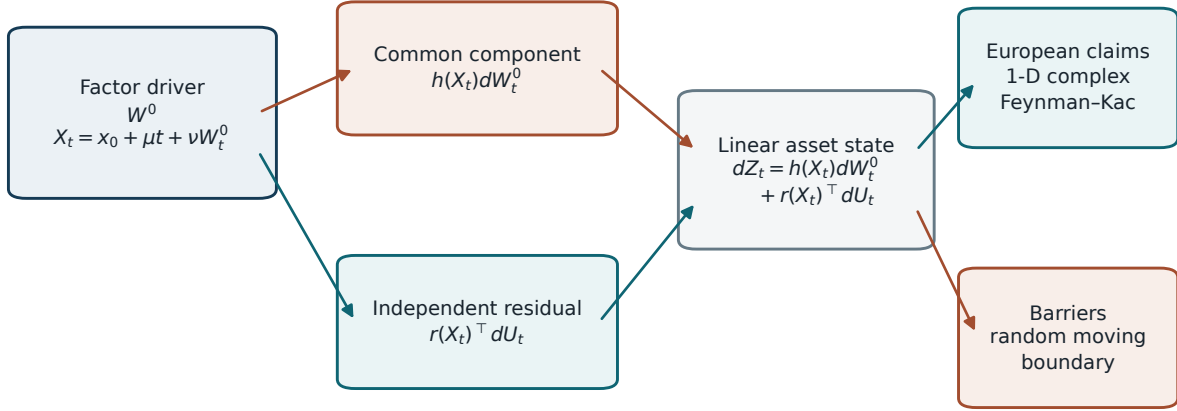


Figure 4: Orthogonal leverage decomposition. Conditioning on the bounded-factor path fixes the common Brownian component, leaving an independent Gaussian residual clock. Terminal European claims retain one-dimensional transform closure, while barriers inherit a random moving boundary.

and covariance

$$\text{Cov}(Z_s, Z_t | X) = \int_0^{s \wedge t} \Gamma(X_r) dr. \quad (42)$$

Proof. The factor SDE Eq. 4 is invertible at path level:

$$W_t^0 = \frac{X_t - x_0 - \mu t}{\nu}.$$

Conditioning on $X_{[0,T]}$ therefore conditions on $W_{[0,T]}^0$. The first term in Eq. 40 is then fixed. The second is an Itô integral against U , which is independent of X , with an integrand that is deterministic under the conditional law. It is a centered Gaussian process with covariance Eq. 42. $\square$

The earlier claim that leverage “destroys conditional Gaussianity” confused centered conditional Gaussianity with conditional Gaussianity itself. Independence is exactly the case $h \equiv 0$, for which the conditional mean is z_0 and the terminal law is a pure variance mixture. Under leverage, the conditional mean is random.

This correction is consistent with established correlated mixing formulas (Willard 1997; Alòs 2006; Das and Langrené 2022). The value here is the explicit closure of the bounded-factor coefficients, not a general discovery that correlated Brownian systems can be orthogonalized.

4.3 An explicit Itô gauge

The stochastic integral in the conditional mean can be converted into an endpoint and an ordinary additive functional. Let $G'(x) = h(x)/\nu$. Since

$$\int f_c(x) dx = \sqrt{x^2 + c^2}, \quad \int s_c(x) dx = c \operatorname{asinh}(x/c),$$

an explicit primitive is

$$G(x) = \frac{1}{\nu} \left[\ell_1 p x + q \ell_1 \sqrt{x^2 + c^2} + q \ell_2 c \operatorname{asinh}(x/c) \right]. \quad (43)$$

Moreover,

$$h'(x) = \frac{qc(\ell_1 c - \ell_2 x)}{(x^2 + c^2)^{3/2}}. \quad (44)$$

Define

$$K(x) = \frac{\mu}{\nu} h(x) + \frac{\nu}{2} h'(x). \quad (45)$$

Proposition 4.1 (Endpoint-additive gauge identity). *For every finite T ,*

$$\int_0^T h(X_t) dW_t^0 = G(X_T) - G(x_0) - \int_0^T K(X_t) dt. \quad (46)$$

Hence

$$Z_T \mid X \sim N \left(z_0 + G(X_T) - G(x_0) - \int_0^T K(X_t) dt, \int_0^T \Gamma(X_t) dt \right). \quad (47)$$

Proof. Apply Itô's formula:

$$dG(X_t) = G'(X_t) dX_t + \frac{1}{2} G''(X_t) \nu^2 dt.$$

Using $G' = h/\nu$, $G'' = h'/\nu$, and $dX_t = \mu dt + \nu dW_t^0$,

$$dG(X_t) = h(X_t) dW_t^0 + \left[\frac{\mu}{\nu} h(X_t) + \frac{\nu}{2} h'(X_t) \right] dt.$$

Integrate and rearrange. $\square$

The inverse-hyperbolic-sine term is the distinctive consequence of allowing the factor to load on the primitive orthogonal driver W^2 . If $\ell_2 = 0$, the primitive contains only a linear term and $\sqrt{x^2 + c^2}$.

4.4 One-dimensional transform closure

Let $Y_t = Z_t - z_0$. The joint generator of (X, Y) is

$$\mathcal{L} = \mu \partial_x + \frac{\nu^2}{2} \partial_{xx} + \frac{Q(x)}{2} \partial_{yy} + \nu h(x) \partial_{xy}. \quad (48)$$

Translation invariance in y is the structural reason that Fourier transformation removes one state dimension.

Theorem 4.2 (One-dimensional leveraged characteristic transform). *For $u \in \mathbb{R}$, define*

$$v(\tau, x; u) = E_{x,0} [e^{iuY_\tau}].$$

Then v is the bounded solution of

$$\partial_\tau v = \frac{\nu^2}{2} v_{xx} + (\mu + iu\nu h(x)) v_x - \frac{u^2}{2} Q(x) v, \quad v(0, x; u) = 1. \quad (49)$$

The terminal characteristic function is

$$\phi_Z(u) = e^{iu z_0} v(T, x_0; u). \quad (50)$$

Equivalently, with the complex potential

$$\mathcal{V}_u(x) = -iuK(x) - \frac{u^2}{2} \Gamma(x), \quad (51)$$

let

$$w(\tau, x; u) = E_x \left[\exp \left(\int_0^\tau \mathcal{V}_u(X_s) ds \right) e^{iuG(X_\tau)} \right].$$

Then

$$\partial_\tau w = \frac{\nu^2}{2} w_{xx} + \mu w_x + \mathcal{V}_u(x) w, \quad w(0, x; u) = e^{iuG(x)}, \quad (52)$$

and

$$\phi_Z(u) = e^{iu(z_0 - G(x_0))} w(T, x_0; u). \quad (53)$$

Proof. For a Fourier mode $e^{iuy}\psi(x)$,

$$\partial_{yy}(e^{iuy}\psi) = -u^2 e^{iuy}\psi,$$

$$\partial_{xy}(e^{iuy}\psi) = iue^{iuy}\psi'(x).$$

Apply the backward Kolmogorov equation for Eq. 48 and factor out e^{iuy} to obtain Eq. 49.

For the second representation, take the conditional characteristic function in Eq. 47:

$$E[e^{iuZ_T} | X] = \exp \left\{ iu \left[z_0 + G(X_T) - G(x_0) - \int_0^T K(X_t)dt \right] - \frac{u^2}{2} \int_0^T \Gamma(X_t)dt \right\}.$$

Taking expectation over X gives Eq. 53, and the usual time-homogeneous Feynman–Kac theorem gives Eq. 52. $\square$

When $\ell = 0$,

$$h = G = K = 0, \quad \Gamma = Q,$$

and both equations reduce to the real-potential variance-clock transform

$$E \left[\exp \left(-\frac{u^2}{2} \int_0^T Q(X_t)dt \right) \right].$$

The transform remains global. Because $|f_c| \leq 1$,

$$0 \leq Q(x) \leq (|p| + |q|)^2.$$

Thus

$$[Y]_T = \int_0^T Q(X_t)dt \leq (|p| + |q|)^2 T.$$

Corollary 4.1 (Sub-Gaussian exponential-moment bound). *For every real θ ,*

$$E[e^{\theta Y_T}] \leq \exp \left(\frac{\theta^2}{2} (|p| + |q|)^2 T \right). \quad (54)$$

Proof. The Doléans exponential

$$\exp\left(\theta Y_T - \frac{\theta^2}{2}[Y]_T\right)$$

is a true martingale because the bracket is bounded. Therefore

$$E[e^{\theta Y_T}] = E\left[\mathcal{E}(\theta Y)_T e^{\theta^2[Y]_T/2}\right] \leq e^{\theta^2(|p|+|q|)^2 T/2} E[\mathcal{E}(\theta Y)_T],$$

and the last expectation equals one. $\square$

Leverage-preserving transform closure is well known in affine and correlated stochastic-volatility models (Heston 1993; Scott 1997; Duffie et al. 2000). Theorem Theorem 4.2 is valuable because the bounded coefficients are nonaffine yet the Fourier reduction and explicit gauge remain one-dimensional.

4.5 Leverage creates skewness

The independent model is a centered variance mixture and therefore symmetric. Under leverage, the random conditional mean generically breaks that symmetry.

Proposition 4.2 (Short-time third moment). *Assume the coefficients are smooth near x_0 . Then*

$$E[Y_T^3] = \frac{3}{2}\nu h(x_0)Q'(x_0)T^2 + o(T^2) = 3pq\nu h(x_0)f'_c(x_0)T^2 + o(T^2). \quad (55)$$

Proof. From Eq. 48,

$$\mathcal{L}(y^3) = 3yQ(x).$$

At $(x_0, 0)$ this is zero. Apply $\mathcal{L}$ once more. Terms carrying a factor y vanish at $y = 0$, and the cross derivative contributes

$$\mathcal{L}^2(y^3)|_{(x_0, 0)} = 3\nu h(x_0)Q'(x_0).$$

The short-time semigroup expansion gives

$$E_{x_0, 0}[Y_T^3] = \frac{T^2}{2}\mathcal{L}^2(y^3)|_{(x_0, 0)} + o(T^2).$$

Finally, $Q'(x) = 2pqf'_c(x)$. $\square$

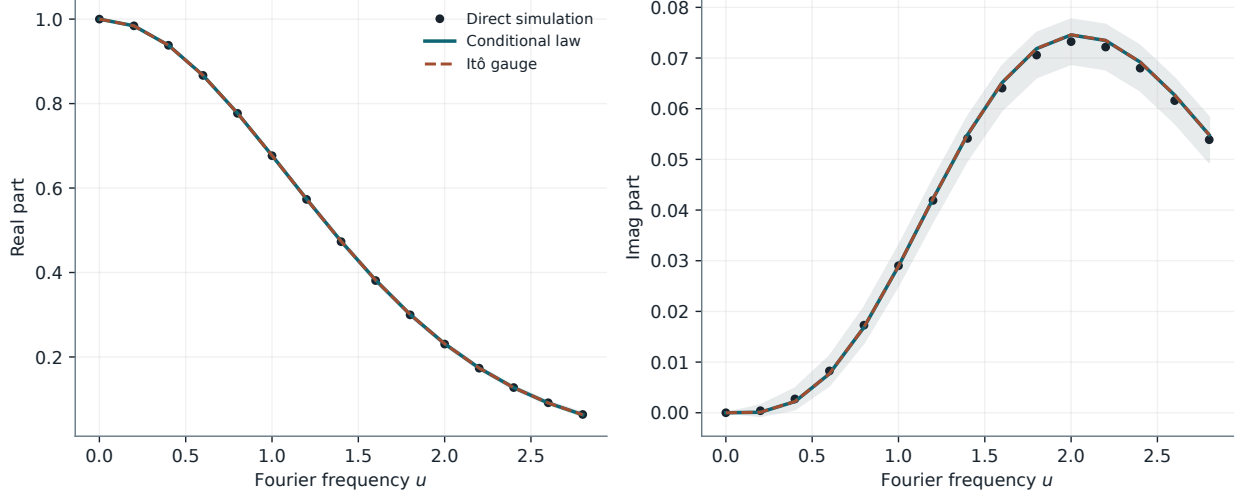


Figure 5: Leveraged characteristic-function validation. Direct simulation, conditional Gaussian integration, and the Itô-gauge representation agree across Fourier frequencies; the shaded band is two Monte Carlo standard errors around the direct estimate.

If $pqh(x_0) \neq 0$, the third moment is nonzero at sufficiently small maturity. This proves loss of exact symmetry but not necessarily a nonzero ATM slope. For a continuous mean-zero terminal law with Bachelier implied standard deviation I ,

$$I'(0) = \frac{1/2 - \mathbb{P}(Y_T > 0)}{\varphi(0)}. \quad (56)$$

A skewed law can still have median zero. The distinction between skewness, median displacement, and ATM slope must therefore be preserved.

The leading coefficient in Eq. 55 is model-specific. The general connection between leverage and option skew is established in stochastic-volatility theory; it is not a new qualitative claim.

4.6 Numerical transform check

The deterministic evidence builder simulates the full correlated Brownian system and compares it with both conditional representations. Across 15 frequencies on $[0, 2.8]$, the maximum direct-versus-conditional gap is 1.63×10^{-3} , below four standard errors. The conditional-versus-gauge gap is 3.87×10^{-5} .

4.7 The true barrier frontier

Define

$$M_t = G(X_t) - G(x_0) - \int_0^t K(X_s) ds, \quad J_t = \int_0^t \Gamma(X_s) ds.$$

On an extension of the conditional probability space,

$$(Z_t \mid X) \stackrel{d}{=} (z_0 + M_t + B_{J_t} \mid X), \quad (57)$$

where B is conditionally independent Brownian motion. For every residual clock, the upper barrier event is

$$\sup_{t \leq T} \{B_{J_t} + M_t\} \geq d - z_0. \quad (58)$$

If $\|\ell\| < 1$ and $Q(X_t) > 0$, then $I_2 - \ell\ell^\top$ is positive definite and J is strictly increasing. In that case Eq. 58 can be rewritten as

$$\sup_{v \leq J_T} \{B_v + M_{J^{-1}(v)}\} \geq d - z_0. \quad (59)$$

If J has flat intervals, Eq. 58 is the canonical representation; no single-valued inverse-clock mean is available in general. The event depends on the entire factor path, not only J_T . Thus the standard terminal scalar-clock reflection derivation does not extend automatically; additional structure is required. When $h \equiv 0$, the moving mean vanishes and the original scalar-clock reduction is recovered.

This boundary is specific. Correlated stochastic-volatility barrier formulas can exist under other multiplicative structures (Rolloos 2022). The open classification problem is to identify nonzero h and strictly increasing residual clocks for which $M_{J^{-1}(v)}$ belongs to a solvable moving- boundary family, together with degenerate-clock cases solvable directly in calendar time.

5 Phase-Gram quotient theory

5.1 Triangular angles and the observable quotient

The parameterization Eq. 11 is an established way to turn unconstrained angles into a valid correlation matrix (Pinheiro and Bates 1996; Rebonato and Jäckel 2000; Rapisarda et al. 2007). Static versions are widely used for optimization, simulation, and prior construction (Joe 2006; Lewandowski et al. 2009; Pourahmadi and Wang 2015; Forrester and Zhang 2020).

The dynamic specification Eq. 13 is deliberately simple: independent unwrapped Brownian angles. The angle lift is Markov on $\mathbb{R}^d$ but has no invariant probability law there. The observable matrix depends only on sines and cosines. It therefore factors through a compact torus.

Write the closed ellipsope as

$$\mathcal{E}_n = \{R \in \mathbb{S}_+^n : \text{diag}(R) = \mathbf{1}\}.$$

Write

$$\Theta_t = (\theta_1(t), \dots, \theta_d(t)) \pmod{2\pi} \in \mathbb{T}^d.$$

The wrapped coordinate process has generator

$$\mathcal{L}_{\mathbb{T}} = \frac{1}{2} \sum_{j=1}^d \sigma_j^2 \partial_{\theta_j}^2.$$

Product Haar measure on $\mathbb{T}^d$ is invariant. “Haar” refers to this coordinate torus; the elliptope of correlation matrices is not a group and has no Haar probability measure.

5.2 Exact determinant factorization

Theorem 5.1 (Phase-Gram determinant and finite-time moments). *For every angle configuration,*

$$\det R_t = \prod_{j=1}^d \sin^2 \theta_j(t). \quad (60)$$

Assume deterministic initial angles. For every positive integer m ,

$$E[(\det R_t)^m] = \prod_{j=1}^d M_{m,j}(t), \quad (61)$$

where

$$M_{m,j}(t) = \frac{1}{4^m} \binom{2m}{m} + \frac{2}{4^m} \sum_{r=1}^m (-1)^r \binom{2m}{m-r} e^{-2r^2 \sigma_j^2 t} \cos(2r\theta_j(0)). \quad (62)$$

For a random initial vector independent of future Brownian increments,

$$E[(\det R_t)^m] = E \left[\prod_{j=1}^d M_{m,j}(t; \theta_j(0)) \right]. \quad (63)$$

Proof. The Gram factor is lower triangular, so

$$\det R = (\det B)^2 = \prod_{i=2}^n B_{ii}^2.$$

By Eq. 11,

$$B_{ii}^2 = \prod_{k < i} \sin^2 \theta_{ik},$$

and every one of the $d = n(n-1)/2$ angles appears once. This proves Eq. 60.

For an integer $m \geq 1$,

$$\sin^{2m} x = \frac{1}{4^m} \binom{2m}{m} + \frac{2}{4^m} \sum_{r=1}^m (-1)^r \binom{2m}{m-r} \cos(2rx). \quad (64)$$

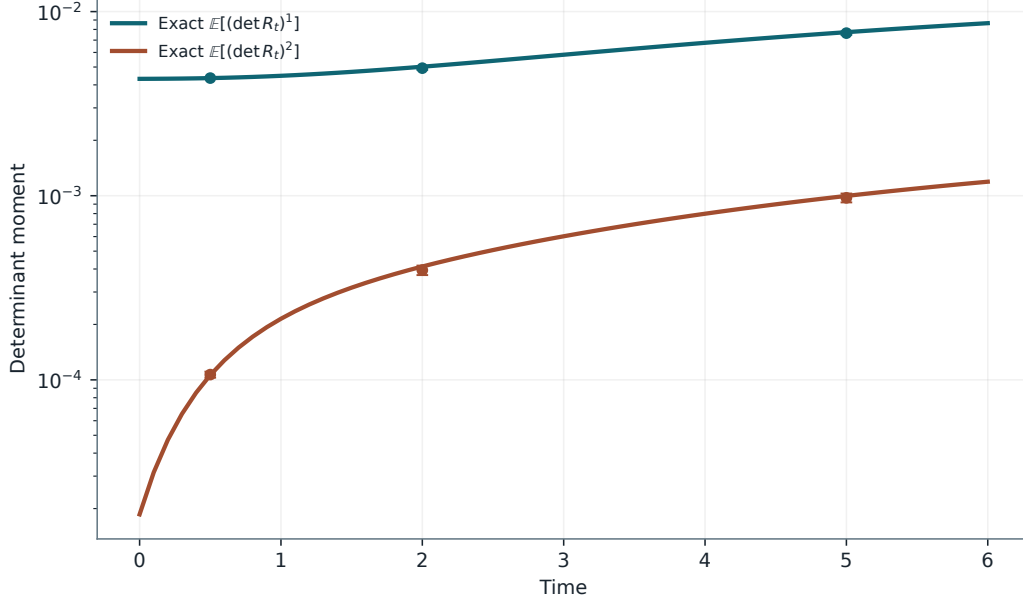


Figure 6: Exact finite-time Phase-Gram determinant moments. Solid curves are the finite Fourier formulas; points with two-standard-error bars are independent Monte Carlo checks.

Since

$$E[\cos(2r\theta_j(t))] = e^{-2r^2\sigma_j^2 t} \cos(2r\theta_j(0)),$$

the expectation of $\sin^{2m} \theta_j(t)$ is Eq. 62. Deterministic initial angles and independent drivers make the terminal angles independent, producing Eq. 61. Conditioning on a random initial vector gives Eq. 63. $\square$ $\square$

The determinant factorization itself is classical in hyperspherical coordinates (Pourahmadi and Wang 2015; Forrester and Zhang 2020), as are product- distribution techniques (Springer and Thompson 1970; Tang and Gupta 1984). The dynamic content in Theorem 5.1 is the explicit finite-time Fourier damping for the Brownian angles.

For $m = 1$,

$$E[\det R_t] = \prod_{j=1}^d \frac{1 - e^{-2\sigma_j^2 t} \cos(2\theta_j(0))}{2}. \quad (65)$$

Figure 6 compares the first two exact moments with independent terminal-angle simulation.

5.3 Compact stationarity

For one coordinate, Brownian motion modulo 2π has transition density

$$p_t(\theta \mid \theta_0) = \frac{1}{2\pi} \left[1 + 2 \sum_{k=1}^{\infty} e^{-k^2 \sigma^2 t/2} \cos(k(\theta - \theta_0)) \right]. \quad (66)$$

The nonconstant Fourier modes decay exponentially. The product process on $\mathbb{T}^d$ therefore converges to product-uniform measure. This is a special case of Brownian and Lévy convergence on compact groups (Applebaum 2014; Liao 2004; Maher 2012).

Theorem 5.2 (Stationary matrix pushforward law). *Let $U_1, \dots, U_d$ be independent uniform random variables on $[0, 2\pi)$ and define*

$$\Pi_n = \text{Law}(R(U_1, \dots, U_d)).$$

Then, for every deterministic initial angle vector,

$$\boxed{R_t \Rightarrow \Pi_n \quad (t \rightarrow \infty).} \quad (67)$$

If the wrapped initial angles are independent uniform variables independent of the Brownian drivers, the matrix process is strictly stationary. Its mean is

$$\boxed{E_{\Pi_n}[R] = I_n.} \quad (68)$$

Proof. Equation Eq. 66 and independence give weak convergence of Θ_t to product Haar measure on $\mathbb{T}^d$. The Gram map is continuous and periodic, so the continuous-mapping theorem gives Eq. 67.

Under product-uniform initialization, the wrapped angle process is stationary, and every finite-dimensional matrix law obtained by applying $R(\cdot)$ is stationary.

For the mean, rows belonging to different asset indices depend on disjoint angle sets and are independent. Every coordinate of a nonfirst row contains either a cosine with zero uniform mean or a product containing at least one sine with zero mean. Hence each nonfirst row has mean zero. Off-diagonal Gram entries have zero expectation; diagonal entries equal one pathwise. $\square$

The conclusion is about matrix marginals and stationary finite-dimensional laws. It does not automatically imply that the image process R_t is Markov on the elliptope: a measurable image of a Markov process need not be Markov. Away from singular fibers the canonical chart is locally invertible; behavior at the boundary is the unresolved point.

The correct long-horizon statement is therefore:

The unwrapped coordinates continue to diffuse, while the observable matrix marginals converge to a compact stationary pushforward law.

The path continues to accumulate quadratic variation and, as shown below, hits the boundary recurrently.

5.4 The invariant law is not uniform and not LKJ

In the canonical interior chart $0 < \theta_{jk} < \pi$, the absolute Jacobian from angles to off-diagonal correlations is

$$J(\theta) = \prod_{k=1}^{n-1} \prod_{j=k+1}^n \sin^{n-k} \theta_{jk}. \quad (69)$$

This Jacobian underlies classical ellipsope-volume and random-correlation calculations (Pourahmadi and Wang 2015; Forrester and Zhang 2020; Eastman et al. 2016; Muirhead 1982). Passing from the unwrapped factor to the unique positive-diagonal canonical chart requires a joint sign argument.

Lemma 5.1 (Flat angles remain flat in the canonical chart). *Off the boundary, let*

$$D_{kk} = \text{sgn}(B_{kk}), \quad L = BD.$$

Then $LL^\top = BB^\top$, L has positive diagonal, and the canonical angles of L are independent uniform variables on $(0, \pi)$ under product-uniform unwrapped angles.

Proof. Since $D^2 = I$, right multiplication by D preserves the Gram matrix and makes $L_{kk} = |B_{kk}| > 0$. Proceed row by row. Conditional on the angles in earlier rows, the column signs $D_{11}, \dots, D_{i-1, i-1}$ are fixed and are independent of the row- i angles. The product-uniform distribution of the hyperspherical row is invariant under every coordinate sign flip: each flip is implemented by a reflection or half-period shift of one of its uniform angles. Multiplication by the fixed earlier column signs therefore preserves the row law. The final sign $D_{ii} = \text{sgn}(B_{ii})$ folds the last coordinate positive; its two equal-measure preimages produce a uniform canonical angle on $(0, \pi)$. Almost everywhere, the map from the row's $[0, 2\pi)^{i-1}$ angles to its canonical $(0, \pi)^{i-1}$ angles has 2^{i-1} equal-measure reflection preimages, so its pushforward density is $\pi^{-(i-1)}$. Induction over i preserves independence of all canonical angles and gives joint density π^{-d} . $\square$

The canonical change-of-variables formula and Lemma 5.1 therefore give the density of Π_n , with respect to Lebesgue measure on the off-diagonal correlations and away from the boundary:

$$f_{\Pi_n}(R) = \pi^{-d} J(\theta(R))^{-1}. \quad (70)$$

It is boundary-singular and depends on the triangular ordering. Here “boundary-singular” means the interior Lebesgue density is unbounded when the boundary is approached; it does not mean that the invariant law charges the boundary. In fact,

$$\Pi_n(\partial\mathcal{E}_n) = 0.$$

By contrast, the LKJ family has

$$f_{\text{LKJ}(\eta)}(R) \propto (\det R)^{\eta-1},$$

whose canonical angle density is proportional to

$$\prod_{k < j} \sin^{2\eta+n-k-2} \theta_{jk}. \quad (71)$$

Flat angles cannot match one value of η for every k once $n \geq 3$. For $n = 2$, flat angles correspond to LKJ(1/2), not the uniform LKJ(1) law (Joe 2006; Lewandowski et al. 2009).

The lack of permutation invariance is already visible at $n = 3$. With

$$R_{12} = \cos \theta_{21}, \quad R_{13} = \cos \theta_{31},$$

$$R_{23} = \cos \theta_{21} \cos \theta_{31} + \sin \theta_{21} \sin \theta_{31} \cos \theta_{32},$$

product-uniform angles give

$$E[R_{12}^2] = E[R_{13}^2] = \frac{1}{2}, \quad E[R_{23}^2] = \frac{3}{8}. \quad (72)$$

Thus $E[R] = I$ hides heterogeneous entry marginals. There is no unique “random correlation matrix” baseline; different generation schemes produce different geometries (Holmes 1991; Bendel and Mickey 1978; Barnard et al. 2000; Archakov et al. 2024).

5.5 Stationary determinant law

Under Π_n ,

$$Y_j = \sin^2 U_j \sim \text{Beta}\left(\frac{1}{2}, \frac{1}{2}\right)$$

independently. Consequently,

$$\boxed{\det R \stackrel{d}{=} \prod_{j=1}^d Y_j.} \quad (73)$$

Corollary 5.1 (Mellin transform and log cumulants). *For $\text{Re } z > -1/2$,*

$$\boxed{E_{\Pi_n}[(\det R)^z] = \left[\frac{\Gamma(z + \frac{1}{2})}{\sqrt{\pi} \Gamma(z + 1)} \right]^d.} \quad (74)$$

Moreover,

$$\boxed{E_{\Pi_n}[\log \det R] = -2d \log 2, \quad \text{Var}_{\Pi_n}(\log \det R) = \frac{d\pi^2}{3}.} \quad (75)$$

In particular,

$$E_{\Pi_n}[\det R] = 2^{-d}, \quad \exp(E_{\Pi_n}[\log \det R]) = 4^{-d}. \quad (76)$$

Proof. For $Y \sim \text{Beta}(1/2, 1/2)$,

$$E[Y^z] = \frac{B(z + \frac{1}{2}, \frac{1}{2})}{B(\frac{1}{2}, \frac{1}{2})} = \frac{\Gamma(z + \frac{1}{2})}{\sqrt{\pi} \Gamma(z + 1)}.$$

Independence yields Eq. 74. Differentiate the log Mellin transform at zero. The digamma identities

$$\psi(1/2) - \psi(1) = -2 \log 2$$

and trigamma identities

$$\psi_1(1/2) - \psi_1(1) = \frac{\pi^2}{3}$$

give Eq. 75. Set $z = 1$ for the arithmetic determinant mean. The geometric mean is the exponential of the log mean. $\square$

These Beta and polygamma facts are standard (Johnson et al. 1995). The notable feature is their exact iid specialization under the flat-angle invariant law. Determinant products and asymptotics also appear in established Gram, Jacobi, and random-correlation ensembles (Rouault 2007; Hanea and Nane 2018); comparisons across ensembles are necessary before declaring one unusually ill-conditioned.

5.6 High-dimensional asymptotics

Let $R_n \sim \Pi_n$ be coupled by one infinite iid sequence of stationary angles and write $d_n = n(n-1)/2$. Each increase in n then adds independent $\text{Beta}(1/2, 1/2)$ determinant factors. The strong law gives

$$\boxed{\frac{1}{d_n} \log \det R_n \longrightarrow -2 \log 2 \quad \text{almost surely.}} \quad (77)$$

The central limit theorem gives

$$\boxed{\frac{\log \det R_n + 2d_n \log 2}{\pi \sqrt{d_n/3}} \Longrightarrow N(0, 1).} \quad (78)$$

Figure 7 shows the finite-dimensional approach to the normal limit. The $n = 3$ standardized law has a visible finite upper endpoint because $\log \det R_n \leq 0$; the boundary moves right as d_n grows.

Determinant decay yields a rigorous but one-sided eigenvalue implication. For positive-definite R_n ,

$$\lambda_{\min}(R_n) \leq (\det R_n)^{1/n}.$$

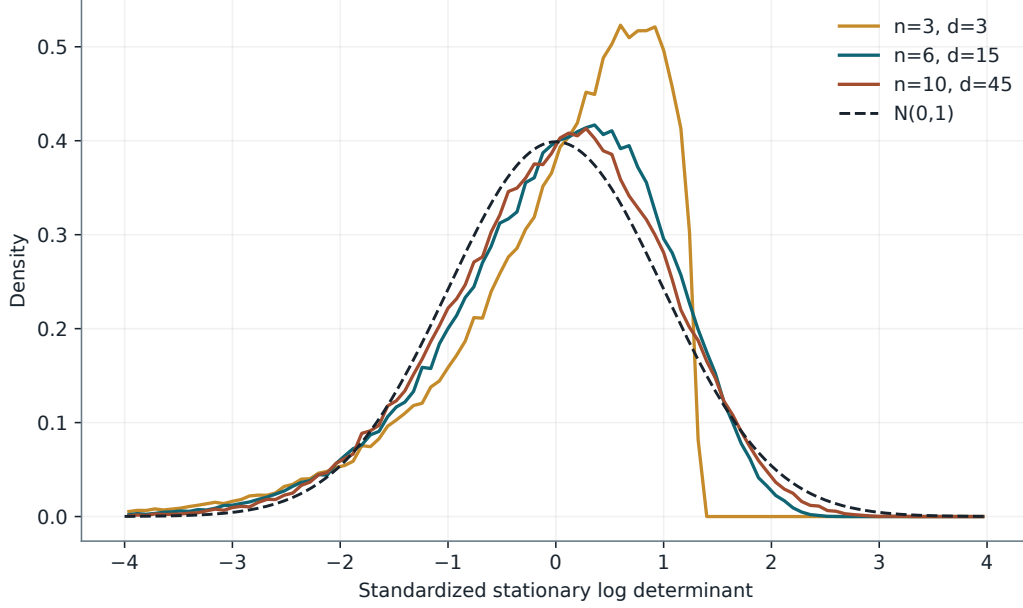


Figure 7: Stationary Phase-Gram log-determinant distributions after exact centering and scaling. The finite-dimensional laws approach the standard normal reference as the number of independent angular factors grows.

Since $2d_n = n(n-1)$, Eq. 77 implies the explicit high-probability bound: for every $\varepsilon > 0$,

$$\mathbb{P}(\lambda_{\min}(R_n) \leq 2^{-(1-\varepsilon)(n-1)}) \rightarrow 1. \quad (79)$$

This is an upper bound, not a smallest-eigenvalue limit. Sharp condition- number and eigenvalue laws remain open.

5.7 First contact with the elliptope boundary

Assume a deterministic interior initial angle vector:

$$x_j = \theta_j(0) \pmod{\pi} \in (0, \pi).$$

The determinant vanishes exactly when at least one angle hits a multiple of π . Let

$$\tau_j = \inf\{t > 0 : \theta_j(t) \notin (k_j\pi, (k_j+1)\pi)\},$$

where the interval contains $\theta_j(0)$, and let

$$\tau_{\partial} = \inf\{t > 0 : \det R_t = 0\} = \min_{1 \leq j \leq d} \tau_j. \quad (80)$$

The killed heat equation on $(0, \pi)$ gives the one-angle survival probability

$$S_j(t) = \frac{4}{\pi} \sum_{m=0}^{\infty} \frac{\sin((2m+1)x_j)}{2m+1} \exp \left[-\frac{1}{2}(2m+1)^2 \sigma_j^2 t \right]. \quad (81)$$

This is the standard odd sine-series solution for Brownian interval exit (Karatzas and Shreve 1991; Borodin and Salminen 2002).

Theorem 5.3 (Exact first-singularity survival law). *For deterministic interior initial angles,*

$$\mathbb{P}(\tau_{\partial} > t) = \prod_{j=1}^d S_j(t). \quad (82)$$

In particular, $\tau_{\partial} < \infty$ almost surely, and the matrix hits the boundary infinitely often.

If the initial angle vector is random but independent of future Brownian increments,

$$\mathbb{P}(\tau_{\partial} > t) = E \left[\prod_{j=1}^d S(t; \theta_j(0) \bmod \pi, \sigma_j) \right]. \quad (83)$$

Proof. Equation Eq. 60 gives

$$\{\tau_{\partial} > t\} = \bigcap_{j=1}^d \{\tau_j > t\}.$$

With deterministic starts, the exit times are independent because the angle drivers are independent. Multiply their survival probabilities. Each one-dimensional Brownian motion exits a finite interval almost surely, so the minimum does also. Brownian recurrence gives infinitely many subsequent hits of $\pi\mathbb{Z}$. Conditioning on a random initial vector gives Eq. 83. $\square$

At every fixed deterministic $t > 0$, the angle distribution is continuous, so the probability of exact singularity is zero and R_t is positive definite almost surely. Pathwise, however, the matrix repeatedly leaves and re-enters the interior. Stationarity and recurrent boundary contact are therefore compatible.

The killed-heat formula Eq. 81 is classical. The model-specific step is the determinant-factorization reduction of the first matrix singularity to the minimum of independent interval exits; no novelty is claimed for the sine-series heat solution itself.

6 Corrected and extended exactness frontiers

6.1 Nonautonomous Feynman–Kac is an ordering problem

Let

$$\mathcal{A}_t = \mathcal{L}_t + \mathcal{V}_t$$

be a time-dependent tilted generator. For a terminal payoff g , define the two-time evolution family

$$Q_{s,t}g(x) = E_{s,x} \left[\exp \left(\int_s^t \mathcal{V}_r(X_r) dr \right) g(X_t) \right]. \quad (84)$$

Under the usual well-posedness, domain, and regularity hypotheses, the standard two-time backward Feynman–Kac theorem remains valid:

$$\boxed{-\partial_s Q_{s,t} = \mathcal{A}_s Q_{s,t}, \quad Q_{t,t} = I.} \quad (85)$$

Time-dependent Feynman–Kac formulas and two-parameter evolution operators are classical (Kac 1949; Karatzas and Shreve 1991; Kato 1953; Pazy 1983; Acquistapace and Terreni 1987; Lunardi 1995). The issue arises when the terminal time is varied while the initial calendar time remains fixed. The correct terminal derivative is a **right action**:

$$\boxed{\partial_t Q_{0,t} = Q_{0,t} \mathcal{A}_t.} \quad (86)$$

It is generally not $\mathcal{A}_t Q_{0,t}$.

Theorem 6.1 (Exact operator-ordering defect). *On a common invariant core on which the displayed products are defined and strongly differentiable,*

$$\boxed{(\partial_t - \mathcal{A}_t)Q_{0,t}g = [Q_{0,t}, \mathcal{A}_t]g = \int_0^t Q_{0,r}[\mathcal{A}_r, \mathcal{A}_t]Q_{r,t}g dr,} \quad (87)$$

where $[B, C] = BC - CB$.

Proof. The first equality is Eq. 86. For the second, fix t and define

$$F(r) = Q_{0,r} \mathcal{A}_t Q_{r,t}.$$

Using

$$\partial_r Q_{0,r} = Q_{0,r} \mathcal{A}_r, \quad \partial_r Q_{r,t} = -\mathcal{A}_r Q_{r,t},$$

we obtain

$$F'(r) = Q_{0,r}(\mathcal{A}_r \mathcal{A}_t - \mathcal{A}_t \mathcal{A}_r) Q_{r,t} = Q_{0,r}[\mathcal{A}_r, \mathcal{A}_t] Q_{r,t}.$$

At the endpoints,

$$F(t) = Q_{0,t} \mathcal{A}_t, \quad F(0) = \mathcal{A}_t Q_{0,t}.$$

Integrate from zero to t . $\square$

Pairwise commutation is sufficient for the naive horizon equation. The integrated identity does not justify a general converse. Time ordering and commutator expansions are central in the Magnus literature (Magnus 1954; Blanes et al. 2009); no novelty is claimed for the operator calculus.

6.1.1 A complete Gaussian diagnostic

Fix $T_* > 0$ and consider deterministic curves satisfying

$$\mu \in L^1([0, T_*]), \quad \sigma \in L^2([0, T_*]),$$

with dynamics

$$dX_t = \mu(t) dt + \sigma(t) dW_t, \quad X_0 = x,$$

and the linear additive functional

$$\mathcal{J}_T = \int_0^T X_s ds.$$

Stochastic Fubini gives

$$\mathcal{J}_T = xT + \int_0^T (T-r)\mu(r) dr + \int_0^T (T-r)\sigma(r) dW_r. \quad (88)$$

Therefore, for real λ ,

$$\boxed{E_x[e^{\lambda \mathcal{J}_T}] = \exp \left\{ \lambda \left[xT + \int_0^T (T-r)\mu(r) dr \right] + \frac{\lambda^2}{2} \int_0^T (T-r)^2 \sigma(r)^2 dr \right\}. \quad (89)}$$

The misordered initial-value equation

$$\partial_T u = \mu(T)u_x + \frac{1}{2}\sigma(T)^2 u_{xx} + \lambda x u, \quad u(0, x) = 1, \quad (90)$$

instead yields

$$\tilde{u}(T, x) = \exp \left\{ \lambda \left[xT + \int_0^T r \mu(r) dr \right] + \frac{\lambda^2}{2} \int_0^T r^2 \sigma(r)^2 dr \right\}. \quad (91)$$

Proposition 6.1 (Constant-coefficient iff for the linear-potential diagnostic). *The transforms Eq. 89 and Eq. 91 agree for every $T \in (0, T_*]$ and every λ if and only if $\mu(t)$ and $\sigma(t)^2$ are constant almost everywhere on $(0, T_*]$.*

Proof. Equality of the coefficients of λ and λ^2 requires, for $a = \mu$ and for $a = \sigma^2$,

$$T \int_0^T a(r) dr = 2 \int_0^T r a(r) dr \quad (92)$$

for every $T \in (0, T_*]$. Let $A(T) = \int_0^T a(r) dr$. Then

$$TA(T) = 2 \int_0^T r a(r) dr.$$

Differentiate almost everywhere:

$$A(T) + Ta(T) = 2Ta(T),$$

so $A(T) = Ta(T)$. Since $A' = a$,

$$A'(T) = \frac{A(T)}{T}$$

almost everywhere on $(0, T_*]$. Hence $A(T) = cT$ and $a(T) = c$ almost everywhere on that interval. Constants plainly satisfy Eq. 92. $\square$

The proper backward solution retains both calendar times:

$$u(s, x; T) = \exp \left\{ \lambda \left[(T-s)x + \int_s^T (T-r)\mu(r) dr \right] + \frac{\lambda^2}{2} \int_s^T (T-r)^2 \sigma(r)^2 dr \right\},$$

and satisfies

$$\partial_s u + \mathcal{A}_s u = 0, \quad u(T, x; T) = 1.$$

Figure 8 displays the ordering and the quantitative divergence for $\mu(t) = t$, $\sigma(t)^2 = 1 + t$, $x = 0.2$, and $\lambda = 0.7$.

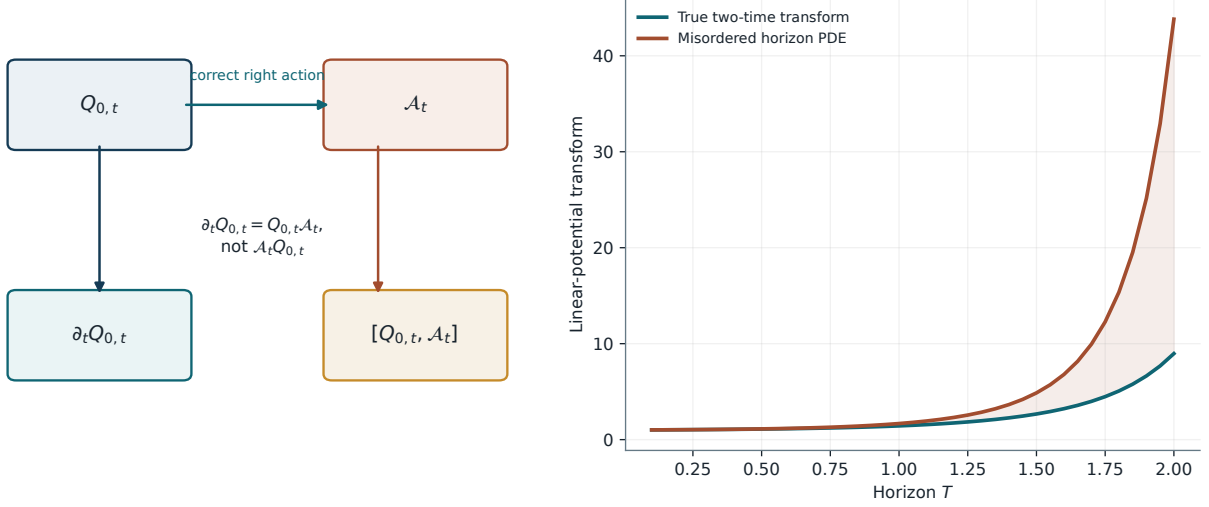


Figure 8: Nonautonomous operator ordering. The left panel distinguishes the correct right action from the invalid left action; the right panel shows the resulting transform error in an exactly solvable Gaussian example.

6.2 Maximal Gaussian scale symmetry

Fix a horizon $[0, T_*]$. Let $F : \mathbb{R} \rightarrow (-1, 1)$ be a homeomorphism and

$$\rho_t = F(X_t/c), \quad c > 0,$$

where

$$X_t = x_0 + \int_0^t \mu(r) dr + \int_0^t \sigma(r) dW_r. \quad (93)$$

Assume deterministic x_0 , $\mu \in L^1[0, T_*]$, and nonnegative deterministic $\sigma \in L^2[0, T_*]$. Equality of path laws means equality as probability measures on $C([0, T_*])$.

Theorem 6.2 (Maximal Gaussian scale equivalence). *Two parameter sets produce the same path law for ρ if and only if*

$$\boxed{\frac{x_0}{c} = \frac{x'_0}{c'}, \quad \frac{\mu(t)}{c} = \frac{\mu'(t)}{c'} \quad a.e., \quad \frac{\sigma(t)^2}{c^2} = \frac{\sigma'(t)^2}{c'^2} \quad a.e.} \quad (94)$$

Under the nonnegative-volatility convention, the last equality is equivalent to $\sigma/c = \sigma'/c'$ almost everywhere.

Proof. Applying F^{-1} to the observed continuous path yields

$$Y_t = \frac{X_t}{c}.$$

This is a Gaussian process with mean

$$m_Y(t) = \frac{x_0}{c} + \int_0^t \frac{\mu(r)}{c} dr$$

and covariance

$$C_Y(s, t) = \int_0^{s \wedge t} \frac{\sigma(r)^2}{c^2} dr.$$

A Gaussian process law is determined by its mean and covariance. Equality of the path laws therefore implies equality of the normalized initial values and of the two integral functions. Absolute continuity then gives the almost- everywhere coefficient equalities in Eq. 94. The converse follows by the same mean/covariance characterization. $\square$

For constant nonnegative coefficients, the positive scale orbit is

$$\boxed{(x_0, c, \mu, \sigma) \mapsto (\kappa x_0, \kappa c, \kappa \mu, \kappa \sigma), \quad \kappa > 0.} \quad (95)$$

Indeed, under the same Brownian path the normalized process is identical pointwise. If signed diffusion coefficients are admitted, an additional measurable sign gauge

$$\sigma'(t) = \kappa \varepsilon(t) \sigma(t), \quad \varepsilon(t) \in \{-1, 1\},$$

preserves the Gaussian law.

The previously written inverse action

$$(c, \mu, \sigma) \mapsto (\kappa c, \mu/\kappa, \sigma/\kappa) \quad (96)$$

is not an invariance. Starting with $c = \sigma = 1$ and $\kappa = 2$ changes normalized volatility from 1 to 1/4, and normalized variance at time t from t to $t/16$.

This theorem is an identifiability statement, not an assertion that scale can never be recovered. A direct observation of X , an externally fixed $\sigma(t)$, or any dimensional measurement tied to the latent coordinate breaks the orbit. One-dimensional scale transformations and their hitting properties are classical (Itô and McKean 1996).

6.3 Joint endpoint–clock law

Let

$$A_t = \int_0^t f_c(X_s) ds$$

for the constant-coefficient Gaussian factor Eq. 4. Define the tilted endpoint kernel

$$K_\xi(t; x, y) dy = E_x [e^{i\xi A_t}; X_t \in dy]. \quad (97)$$

It satisfies the forward equation

$$\partial_t K_\xi = -\mu \partial_y K_\xi + \frac{\nu^2}{2} \partial_{yy} K_\xi + i\xi f_c(y) K_\xi, \quad K_\xi(0; x, y) = \delta_x(y). \quad (98)$$

This is the standard additive-functional/Feynman–Kac construction (Kac 1949). Retaining y rather than integrating it out gives the joint endpoint–clock transform without introducing a second spatial PDE coordinate.

Theorem 6.3 (Smooth endpoint–clock density). *Assume the joint diffusion is nonexplosive and its Stratonovich vector fields are C^∞ with bounded derivatives of every order. For every $T > 0$, (A_T, X_T) has a smooth density. In the distributional Fourier sense,*

$$p_{A_T, X_T}(a, y) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-i\xi a} K_\xi(T; x_0, y) d\xi. \quad (99)$$

Consequently,

$$p_{A_T, \rho_T}(a, r) = \frac{c}{(1 - r^2)^{3/2}} p_{A_T, X_T} \left(a, \frac{cr}{\sqrt{1 - r^2}} \right). \quad (100)$$

Proof. In coordinates (x, a) , write the joint SDE in Stratonovich form. Since the diffusion coefficient is constant, the Itô and Stratonovich drifts coincide:

$$V_1 = \nu \partial_x, \quad V_0 = \mu \partial_x + f_c(x) \partial_a.$$

Their bracket is

$$[V_1, V_0] = \nu f'_c(x) \partial_a.$$

At every finite x ,

$$f'_c(x) = \frac{c^2}{(x^2 + c^2)^{3/2}} > 0.$$

Thus V_1 and $[V_1, V_0]$ span both tangent directions. The parabolic, or weak, Hörmander condition holds, so the joint law has a smooth density under the stated global hypotheses. Fourier transformation in a gives Eq. 98, and inverse Fourier transformation gives Eq. 99. The last formula is the monotone Jacobian Eq. 3. $\square$

The bracket argument belongs to classical hypoelliptic theory (Kolmogoroff 1934; Hörmander 1967; Malliavin 1978; Kusuoka and Stroock 1985; Norris 1986; Nualart 2006; Hairer 2011; Delarue and Menozzi 2010). It is not uniform: $f'_c(x) \rightarrow 0$ as $|x| \rightarrow \infty$. The theorem claims smoothness, not global Gaussian lower bounds or everywhere strict positivity.

Pointwise use of Eq. 99 requires additional Fourier decay, such as

$$\xi \mapsto K_\xi(T; x_0, y) \in L^1(\mathbb{R}).$$

Without it, distributional inversion is the correct statement.

For a corridor indicator

$$O_T = \int_0^T \mathbf{1}_{\{\ell < X_s < h\}} ds,$$

the geometry is different. The integrand is constant on open regions, so paths can remain entirely outside or inside the corridor. Depending on the initial point, the joint law may contain singular measures tied to killed endpoint kernels: an atom at $O_T = T$ is available when $x_0 \in (\ell, h)$, while an atom at $O_T = 0$ is available when $x_0 \notin [\ell, h]$. A boundary start can have neither because of Brownian regularity. Any present singular components must be separated before numerical inversion.

6.4 Spectrally one-sided Lévy barriers

Let X be a spectrally negative Lévy process that is not the negative of a subordinator. Write $x = X_0$ and define its Laplace exponent by

$$\psi(\theta) = \log E[e^{\theta(X_1 - X_0)}], \quad \theta \geq 0.$$

For $\delta \geq 0$, let $\Phi(\delta)$ be the largest nonnegative solution of

$$\psi(\Phi(\delta)) = \delta.$$

Use the standard strict passage times

$$\tau_b^+ = \inf\{t > 0 : X_t > b\}, \quad \tau_a^- = \inf\{t > 0 : X_t < a\}. \quad (101)$$

The δ -scale function is zero on the negative half-line and is normalized by

$$W^{(\delta)}(x) = 0, \quad x < 0,$$

$$\boxed{\int_0^\infty e^{-\theta y} W^{(\delta)}(y) dy = \frac{1}{\psi(\theta) - \delta}, \quad \theta > \Phi(\delta).} \quad (102)$$

Define

$$Z^{(\delta)}(x) = 1 + \delta \int_0^x W^{(\delta)}(y) dy. \quad (103)$$

These conventions and the following identities are standard in spectrally negative fluctuation theory (Bertoin 1996; Doney 2007; Kyprianou 2014; Kuznetsov et al. 2013).

Let $\rho_t = f_c(X_t)$ and $r^* > \rho_0$. Set

$$b = f_c^{-1}(r^*).$$

Strict monotonicity gives the pathwise identity

$$\inf\{t > 0 : \rho_t > r^*\} = \tau_b^+. \quad (104)$$

Proposition 6.2 (One-sided bounded-factor fluctuation identities). *For $x < b$,*

$$\boxed{E_x[e^{-\delta\tau_b^+}; \tau_b^+ < \infty] = e^{-\Phi(\delta)(b-x)}}. \quad (105)$$

For $a < x < b$,

$$\boxed{E_x[e^{-\delta\tau_b^+}; \tau_b^+ < \tau_a^-] = \frac{W^{(\delta)}(x-a)}{W^{(\delta)}(b-a)}}, \quad (106)$$

and

$$\boxed{E_x[e^{-\delta\tau_a^-}; \tau_a^- < \tau_b^+] = Z^{(\delta)}(x-a) - Z^{(\delta)}(b-a) \frac{W^{(\delta)}(x-a)}{W^{(\delta)}(b-a)}}. \quad (107)$$

Proof. The formulas for the driver are the standard one- and two-sided exit identities. Equation Eq. 104 transports them pathwise to the bounded factor. No stochastic-calculus correction appears because the argument concerns order of sample paths, not a differential. $\square$

The theory has a long development, including early resolvent constructions, reflected exits, creeping, and scale-function regularity (Suprun 1976; Avram et al. 2004; Alili and Kyprianou 2005; Chan et al. 2011). Occupation and joint occupation/first-passage transforms already exist (Landriault et al. 2011; Loeffen et al. 2014), as does state-dependent Omega killing (Li and Palmowski 2018).

Accordingly, Proposition 6.2 is a correction, not a novelty theorem: reflection dies under jumps, but exact barriers do not die universally. For two-sided VG or NIG drivers, this one-scale-function

simplification does not apply directly; Wiener–Hopf or numerical fluctuation methods may remain available.

A sharper open problem begins from the existing generalized Omega-scale Volterra equations and the bounded, sign-changing complex potential

$$\omega_\xi(x) = \delta - i\xi f_c(x).$$

The task is to prove the corresponding complex-valued two-sided exit and resolvent identities rigorously, establish real-frequency decay or L^1_ξ conditions, and construct a stable joint barrier–clock inversion. Analytic continuation may be useful, but it is not itself the required endpoint.

7 Unified implications and limitations

7.1 Exactness map

The paper’s central distinction is between the failure of one representation and the failure of tractability itself.

Table 3: Corrected exactness frontier. “Exact” includes a closed transform or deterministic equation when no elementary density is claimed.

Setting	Static law	Terminal European transform	Centered smile symmetry	Scalar terminal-clock barrier	Stationary observable law	Exact fixed-level factor barrier
Independent Gaussian factor	exact push-forward	1-D real-potential FK	yes	yes for driftless linear combinations	no for scalar Brownian state	reflection principle
Leveraged Gaussian factor	exact push-forward	1-D complex FK	generically no	not from J_T alone	no for scalar Brownian state	reflection for the factor itself
Brownian Phase-Gram angles	exact Gaussian-angle pushforward	finite-dimensional angle integral	not applicable	not applicable	yes, compact pushforward	exact determinant-boundary survival
Nonautonomous Gaussian factor	exact finite-dimensional Gaussian law	two-time backward or forward tilted PDE	under centered conditional law	special time-change classes only	coefficient-dependent	curved-boundary problem
Spectrally negative Lévy factor	exact transform pushforward	1-D PIDE/transform	driver-dependent	no Brownian reflection	driver-dependent	fixed-level scale-function identities

Setting	Static law	Terminal European transform	Centered smile symmetry	Scalar terminal-clock barrier	Stationary observable law	Exact fixed-level factor barrier
Two-sided Lévy factor	exact transform pushforward	1-D PIDE/transform	driver-dependent	no	driver-dependent	no one-scale simplification

7.2 Correction ledger

Table 4: Mathematical correction ledger for the bounded-factor program.

Earlier statement	Correct statement
ATM implied correlation equals the mean clock.	It equals the mean clock minus the exact square-root-variance Jensen term in Eq. 26.
Leverage destroys conditional Gaussianity.	It introduces a random conditional mean; Gaussianity and 1-D European transform closure survive.
Brownian Phase-Gram has no stationary law.	Unwrapped angles have no invariant law on $\mathbb{R}^d$; matrix marginals converge to a compact stationary pushforward.
The scale orbit uses inverse drift/volatility scaling.	All dimensional latent parameters scale in the same direction as Eq. 95.
Backward Feynman–Kac fails under calendar-time coefficients.	The two-time theorem remains valid; a one-parameter left operator action is misordered.
Lévy barriers are dead.	Brownian reflection is dead; one-sided scale-function barriers remain exact.

7.3 Limitations

1. The implied-curvature theorem is local at the money. Global wings require the full mixture law.
2. The leveraged terminal transform is one-dimensional, but its potential is complex and strike inversion remains numerical.
3. The moving-boundary obstruction does not rule out exceptional leveraged barrier families.
4. The stationary Phase-Gram law is ordering-dependent, boundary-singular, and not LKJ for $n \geq 3$.
5. Stationarity of the matrix image does not prove a global Markov quotient generator at singular fibers.
6. Determinant asymptotics give only an upper bound on the smallest eigenvalue.
7. The endpoint–clock Hörmander condition is pointwise but not uniform.
8. Fourier inversion of the joint kernel is distributional unless frequency decay is established.
9. One-sided Lévy scale functions do not directly solve two-sided jump drivers.

10. The nonautonomous commutator formula requires a common invariant core and strong differentiability; unbounded-domain questions are real, not notation.

7.4 Open problems

1. **Leveraged barrier classification.** Characterize every nonzero leverage configuration for which the conditional calendar-time boundary is solvable, including strict clocks for which $M_{J^{-1}(v)}$ is affine and degenerate clocks requiring no inverse.
2. **Higher smile derivatives.** Determine the complete ATM derivative hierarchy and which positive or negative moments of S it identifies.
3. **Mixture identifiability.** Decide when one full normal-implied smile uniquely determines the centered mixing distribution.
4. **Phase-Gram quotient dynamics.** Prove or disprove Markov lumpability of the matrix image, especially through boundary fibers.
5. **Spectral conditioning.** Derive sharp smallest-eigenvalue and condition- number asymptotics under Π_n and compare them with LKJ and Gram baselines.
6. **Leveraged endpoint–clock kernels.** Develop a stable joint transform for the endpoint and two additive functionals created by the Itô gauge.
7. **Complex Omega killing.** Starting from generalized Omega-scale Volterra equations, prove complex two-sided exit/resolvent identities for $\delta - i\xi f_c(x)$, establish real-frequency decay, and construct a stable inversion.
8. **Observation channels and scale.** Classify which state-space observation schemes preserve or break Eq. 94.

8 Conclusion

Bounded-factor tractability is not exhausted by independence, unwrapped stationarity, time-homogeneous notation, or the Brownian reflection principle. Each of those assumptions buys a particularly simple representation. Removing one can destroy that representation without destroying every exact structure.

For centered Gaussian variance mixtures, the ATM Bachelier level is $E[\sqrt{V}]$, not $\sqrt{E[V]}$. In correlation units this distinction is the exact Jensen wedge Eq. 26, and clock dispersion pushes spread and basket implied correlations in opposite directions. Under Brownian leverage, conditioning produces a random mean and a residual clock. The explicit bounded-factor Itô gauge and Fourier reduction preserve a one-dimensional European transform, while the short-time third moment quantifies the resulting asymmetry. For Brownian triangular angles, periodic quotienting produces a stationary matrix law with exact determinant statistics and recurrent boundary contact, even though the unwrapped angle lift diffuses forever.

The supporting corrections replace broad failure statements by precise frontiers: simultaneous scale symmetry, two-time nonautonomous evolution, hypoelliptic endpoint–clock smoothing, and one-sided Lévy fluctuation identities.

The central closure principle is therefore

bounded factor + Brownian leverage + linear asset combination with terminal European payoff $\implies$ one-dimension
--

This principle is not universal: barriers retain the random mean path, matrix stationarity is not automatically Markovianity, and two-sided jumps exceed one-scale fluctuation theory. Its value is exactly that it states what survives, what fails, and why.

9 Technical appendices

9.1 Appendix A: Bachelier differentiation ledger

Let $d = k/s$. Starting from

$$\mathcal{B}(s, k) = s\varphi(d) - k\Phi(-d),$$

the derivatives used in Theorem 3.1 are:

$$\partial_s[s\varphi(d)] = \varphi(d) + s\varphi'(d) \left(-\frac{k}{s^2}\right) = \varphi(d) + d^2\varphi(d),$$

$$\partial_s[-k\Phi(-d)] = -k\varphi(d) \frac{k}{s^2} = -d^2\varphi(d),$$

so $\mathcal{B}_s = \varphi(d)$. Likewise,

$$\partial_k[s\varphi(d)] = \varphi'(d) = -d\varphi(d),$$

$$\partial_k[-k\Phi(-d)] = -\Phi(-d) + d\varphi(d),$$

so $\mathcal{B}_k = -\Phi(-d)$ and

$$\mathcal{B}_{kk} = \frac{\varphi(d)}{s}.$$

The remaining derivatives at $k = 0$ are

$$\mathcal{B}_{sk}(s, 0) = 0, \quad \mathcal{B}_{ss}(s, 0) = 0.$$

These identities are sufficient for the implicit-function calculation; no closed-form inversion away from ATM is used.

9.2 Appendix B: Brownian covariance algebra

With the order (W^1, W^2, W^0) ,

$$\det \Sigma_W = \begin{vmatrix} 1 & 0 & \ell_1 \\ 0 & 1 & \ell_2 \\ \ell_1 & \ell_2 & 1 \end{vmatrix} = 1 - \ell_1^2 - \ell_2^2.$$

Let the driver-loading matrix from (W^1, W^2, W^0) to (F_1, F_2, X) be

$$L(x) = \begin{pmatrix} \sigma_1 & 0 & 0 \\ \sigma_2 f_c(x) & \sigma_2 s_c(x) & 0 \\ 0 & 0 & \nu \end{pmatrix}.$$

Then the covariance-rate matrix is

$$\Sigma_{F,X}(x) = L(x) \Sigma_W L(x)^\top$$

and

$$\det \Sigma_{F,X}(x) = (\det L(x))^2 \det \Sigma_W = \sigma_1^2 \sigma_2^2 \nu^2 s_c(x)^2 (1 - \ell_1^2 - \ell_2^2).$$

For the linear combination, $g = (g_1, g_2)^\top$ and

$$dZ_t = g(X_t)^\top (\ell dW_t^0 + C dU_t).$$

Therefore

$$h = \ell^\top g, \quad \Gamma = g^\top C C^\top g = g^\top (I - \ell \ell^\top) g = Q - h^2.$$

9.3 Appendix C: Generator expansion for skewness

For a sufficiently regular test function $\phi(x, y)$,

$$\mathcal{L}\phi = \mu \phi_x + \frac{\nu^2}{2} \phi_{xx} + \frac{Q(x)}{2} \phi_{yy} + \nu h(x) \phi_{xy}.$$

Take $\phi = y^3$. Then

$$\phi_x = \phi_{xx} = \phi_{xy} = 0, \quad \phi_{yy} = 6y,$$

so

$$\mathcal{L}\phi = 3yQ(x).$$

For $\psi(x, y) = 3yQ(x)$,

$$\psi_x = 3yQ'(x), \quad \psi_{xx} = 3yQ''(x), \quad \psi_{yy} = 0, \quad \psi_{xy} = 3Q'(x).$$

At $(x_0, 0)$ only the mixed term survives:

$$\mathcal{L}\psi(x_0, 0) = 3\nu h(x_0)Q'(x_0).$$

Under a local semigroup expansion,

$$P_T\phi = \phi + T\mathcal{L}\phi + \frac{T^2}{2}\mathcal{L}^2\phi + o(T^2),$$

which proves Eq. 55. Bounded Q supplies all terminal moments; the expansion still requires the stated local regularity.

9.4 Appendix D: Phase-Gram Fourier and Beta identities

Euler's formula gives

$$\sin x = \frac{e^{ix} - e^{-ix}}{2i}.$$

Raising to the even power $2m$, pairing conjugate modes, and collecting the zero mode yields Eq. 64. Gaussian-angle damping follows from

$$E[e^{ik(\theta_0 + \sigma W_t)}] = e^{ik\theta_0 - k^2\sigma^2 t/2}.$$

For U uniform on $[0, 2\pi)$ and $Y = \sin^2 U$, the two preimages in each half-period give

$$f_Y(y) = \frac{1}{\pi\sqrt{y(1-y)}}, \quad 0 < y < 1,$$

the Beta(1/2, 1/2) density. Its logarithmic cumulant generating function is

$$\log E[Y^z] = \log \Gamma(z + \tfrac{1}{2}) - \log \Gamma(z + 1) - \frac{1}{2} \log \pi.$$

Differentiation at zero gives the digamma and trigamma expressions used in Eq. 75.

9.5 Appendix E: Killed-heat derivation of the singularity time

For one angle shifted into $(0, \pi)$, the survival function solves

$$\partial_t u = \frac{\sigma^2}{2} u_{xx}, \quad u(t, 0) = u(t, \pi) = 0, \quad u(0, x) = 1.$$

Expand the constant initial condition in the Dirichlet sine basis:

$$1 = \sum_{k=1}^{\infty} b_k \sin(kx), \quad b_k = \frac{2}{\pi} \int_0^{\pi} \sin(kx) dx.$$

Thus $b_k = 0$ for even k and

$$b_{2m+1} = \frac{4}{\pi(2m+1)}.$$

The heat semigroup multiplies the k th mode by $e^{-k^2 \sigma^2 t/2}$, producing Eq. 81. Independence and the determinant product then give Eq. 82.

9.6 Appendix F: Evolution-family commutator derivation

The derivation of Eq. 87 uses two different strong derivatives:

$$\partial_r Q_{0,r} = Q_{0,r} \mathcal{A}_r$$

and

$$\partial_r Q_{r,t} = -\mathcal{A}_r Q_{r,t}.$$

They act on different sides because one changes the terminal time and the other changes the initial time. This is the operator-level version of the different weights r and $T-r$ in the Gaussian diagnostic. Domain invariance is indispensable when $\mathcal{A}_t$ is unbounded.

9.7 Appendix G: Gaussian path-law identifiability

If two normalized Gaussian processes have the same law on $C([0, T_*])$, their finite-dimensional means and covariances agree. Taking one time identifies the normalized mean function; taking pairs identifies the integrated normalized variance. Conversely, equality of those two functions gives equality of every finite-dimensional Gaussian law. Continuity then identifies the path measure.

The theorem assumes a known common homeomorphism F . If F itself carries unknown scale or shape parameters, additional group actions can appear and must be analyzed separately.

9.8 Appendix H: Hypoelliptic and Lévy conventions

The Hörmander bracket is computed for Stratonovich fields. Since the sole diffusion field $V_1 = \nu \partial_x$ is constant, the Itô-to-Stratonovich correction vanishes. The pointwise span is two-dimensional, but the determinant of the spanning fields is proportional to $f'_c(x)$ and tends to zero at infinity.

For a spectrally negative Lévy process, $W^{(\delta)}$ is continuous on $[0, \infty)$ under the standing spectrally negative hypothesis, but further differentiability depends on the Gaussian component and Lévy measure (Chan et al. 2011). None of the exit formulas used in the main text requires unqualified smoothness of the scale function.

Strict passage conventions match Kuznetsov et al. (2013). If non-strict passage is desired, its equivalence must be checked against regularity and creeping for the chosen process; the paper does not silently interchange the two.

9.9 Appendix I: Numerical evidence ledger

Table 5: Selected deterministic and fixed-seed validation anchors.

Check	Value
Clock density mass	1.000000000000
$E[A_T]$	0.106628721753
$\text{Var}(A_T)$	0.111738386500
Spread ATM implied correlation	0.13453
Basket ATM implied correlation	0.08201
Clock-density mean error versus independent quadrature	-1.45×10^{-6}
Clock-density variance error versus independent quadrature	8.81×10^{-7}
Maximum direct/conditional leveraged CF gap	0.001627827791
Maximum conditional/gauge CF gap	3.8689×10^{-5}
Nonautonomous transform gap at $T = 1.3$	0.888672528169

The evidence generator uses Python, NumPy, SciPy, and Matplotlib in the existing project environment. It writes every plotted number to CSV or JSON, uses seed 20260822, batches the leveraged simulation, and checks Phase-Gram moments against independent terminal-angle simulation. The exact formulas in the manuscript remain the canonical statements.

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